Appendix

Contents

I Dataset Processing Details 2
- I-A Dataset Selection . . . 2
- I-B DMD Split and Training Data Construction . . . 3

II Evaluation Metrics 6
- II-A Detection Performance Metrics . . . 6
- II-B Deployment Efficiency Metrics . . . 8

III Baseline 9
- III-A Baseline Selection . . . 9

IV Implementation Details 11
- IV-A Edge-side Event Mining Parameters . . . 11
- IV-B RTMPose-based Hand Box Tool . . . 13
- IV-C Agent Prompt Design . . . 14
- IV-D Edge Devices and Runtime Environment . . . 15

V Additional Experimental Analysis 17
- V-A Interaction between Annotation Models and Edge Detectors . . . 17
- V-B Impact of Reasoning Mode on Downstream Detector Training . . . 21
- V-C Class-level Sensitivity to Pseudo-label Quality . . . 23
- V-D RGB–IR Differences in Class-wise Detection . . . 24
- V-E Detailed Results of Multi-round Detector Training . . . 25

References 27

Live Demo. We provide a live demonstration of our system at: https://DataDet.github.io/.

I. Dataset Processing Details

A. Dataset Selection

DataDet is designed for continuous in-vehicle driver monitoring rather than isolated image-level distracted-driving classification. Therefore, the evaluation dataset should provide continuous in-cabin video streams that allow the edge-side mining module to scan raw frames and select high-risk candidate events. It should also contain both normal-driving and distracted-driving periods, because event mining is meaningful only when distracted behaviors are sparse and mixed with redundant normal frames. Reliable annotations are also needed for quantitative evaluation. These properties are essential for evaluating the full DataDet pipeline, including edge-side event mining, cloud-side pseudo-label construction, model training, and edge deployment.

We considered several public distracted-driving and driver-monitoring datasets. Existing datasets can be roughly divided into three categories. The first category contains image-level datasets, such as AUC. These datasets are useful for conventional distracted-driving image classification, but they do not provide continuous video streams. Therefore, they cannot support the evaluation of edge-side event mining, upload reduction, or closed-loop video-based data construction.

The second category contains datasets with image frames and short video clips, such as 100 Driver. These datasets provide more temporal information than image-only datasets, but many clips are short and mainly centered around distracted-driving behaviors. As a result, they contain limited normal-driving context and are less suitable for testing whether a mining module can discover sparse high-risk events from long raw driving streams.

The third category contains longer in-cabin activity videos, such as Drive&Act. Although these datasets provide richer temporal recordings, their recording viewpoints, occlusions, and action settings are not always suitable for standard distracted-driving detection. In many cases, key visual evidence is incomplete due to hand/body occlusion, indirect viewpoints, or complex interaction layouts. As a result, it is difficult to make deterministic judgments about whether a specific distracted-driving behavior truly occurs. Since DataDet relies on reliable event selection and pseudo-label verification, using such data as the main benchmark may introduce ambiguity that is unrelated to the proposed method itself.

In contrast, DMD better matches the practical requirements of continuous in-vehicle

driver monitoring. It provides long continuous in-cabin video streams and real-car side-view recordings with clearer evidence for driver behaviors and in-cabin interactions. These properties make it possible to evaluate not only final recognition accuracy, but also the complete process of discovering useful events from raw videos, constructing reliable training data, and deploying lightweight models under realistic data and resource constraints. Therefore, DMD is selected as the main benchmark because it provides more complete support for evaluating data construction and deployment in practical distracted-driving detection scenarios.

Dataset Type	Representative Dataset	Data Form	Viewpoint	Normal-driving Videos
Image-level datasets	AUC	Images	Clear single-frame views	No
Short-clip driver datasets	100 Driver	Images , short videos	Generally usable, but clip-level context is limited	Limited
Long in-cabin activity datasets	Drive&Act	Long videos	Often affected by occlusion and complex viewpoints	Partially available
Continuous driver-monitoring datasets	DMD	Long in-cabin videos	Real-car side-view with relatively complete driver evidence	Available

TABLE I: Comparison of different dataset types for evaluating continuous distracted-driving detection. DMD is selected as the main benchmark because it provides long in-cabin videos, relatively complete visual evidence, and normal-driving context needed for practical driver-monitoring evaluation.

It should be noted that selecting DMD does not mean that other datasets are unimportant. Image-level datasets, short-clip driver datasets, and long in-cabin activity datasets are valuable for different purposes, such as distracted-driving image classification, short-term driver action recognition, and general in-cabin activity understanding. However, the main objective of DataDet is to evaluate whether unlabeled continuous driving videos can be converted into high-quality pseudo-labeled training data under bandwidth and edge-computation constraints. For this purpose, DMD provides a better match to the required evaluation protocol than the other available datasets.

B. DMD Split and Training Data Construction

We use the distraction-related part of the Driver Monitoring Dataset (DMD) as the main experimental source. DMD provides in-cabin driver monitoring videos organized by participant groups, subjects, recording sessions, camera views, and modalities. In this work, we focus on the real-car distraction recordings captured from the side-view body camera, because this view provides relatively complete visual evidence of the driver's upper body, hands, steering wheel, and interactions with in-cabin objects. These visual cues are necessary for evaluating fine-grained distracted-driving categories such as radio operation, drinking, phone calling, texting, and hand-on-wheel states.

The DMD material is organized into group folders from gA to gZ. Each group contains several participants, and each participant contains recording sessions such as s1, s2, and

Split	Groups	Videos / Modality	Data Source	Usage
Training mining	gA–gE	71	Continuous real-car side-view videos	Edge mining and cloud pseudo-label construction
Testing	gF–gZ	78	Manually annotated real-car side-view videos	Final quantitative evaluation

TABLE II: DMD split used in our experiments. Training candidates and test samples come from disjoint group folders and videos.

s3. The videos and corresponding OpenLABEL annotation files are stored under each session. The simplified directory structure used in our experiments is shown below:

```
DMD/
  gA/
    subject_id/
      s1/
        *_rgb_body.mp4
        *_ir_body.mp4
        *_rgb_ann_distraction.json
      s2/
        *_rgb_body.mp4
        *_ir_body.mp4
        *_rgb_ann_distraction.json
      s3/
        ...
  gB/
    ...
  ...
  gZ/
    ...
```

To avoid temporal leakage, the mined training videos and the test videos are completely separated at the group and video levels. Specifically, DataDet mines training candidates from gA–gE, corresponding to 71 videos per modality. The test set is constructed from gF–gZ, corresponding to another 78 videos per modality and about 100K manually annotated frames. RGB and IR streams are synchronously recorded and follow the same group-level split. Therefore, there is no overlap between training and testing videos, no adjacent-frame leakage, and no event-level leakage.

For training data construction, the edge-side mining module first scans the continuous training videos and extracts about 50K candidate frames from each modality for cloud-side processing. These frames are not directly used as the final training set. Instead, they are sent to the cloud annotation pipeline, where the agent generates hand-centric pseudo labels, verifies uncertain cases, and filters unreliable samples. After this stage, we further perform class-balanced sampling to construct the final YOLO training set, resulting in about 9.3K RGB frames and 6.7K IR frames.

The reduction from 50K mined candidates to the final training set does not mean that most mined frames are invalid. As shown in the mining-quality experiment, the

edge-side mining module already enriches high-risk frames effectively. The smaller final training set is mainly caused by class balancing. In the mined candidate pool, different distracted-driving categories naturally appear with different frequencies. If all verified samples are directly used for training, frequent classes such as hand-on-wheel states may dominate the training set and suppress minority behaviors such as drinking, texting, or hair and makeup. To avoid this imbalance, we perform multi-label instance-balanced sampling on the training split.

Each image may contain two hand labels at the same time, so we count class instances rather than only image-level labels. Let $\mathcal{C}$ denote the set of distracted-driving classes and let n_c be the number of available instances for class c. We first compute a base target T from the non-wheel classes:

$$T = \max\left(T_{\min}, \operatorname{mean}\{n_c \mid c \in \mathcal{C}_{\text{non-wheel}}\}\right), \tag{1}$$

where $T_{\min}$ is the minimum target number of instances per class. For wheel-related classes, we allow a slightly larger target because these states naturally appear more often:

$$T_c = \begin{cases} \min(\gamma T, n_c), & c \in \mathcal{C}_{\text{wheel}}, \\ \min(T, n_c), & c \in \mathcal{C}_{\text{non-wheel}}, \end{cases} \tag{2}$$

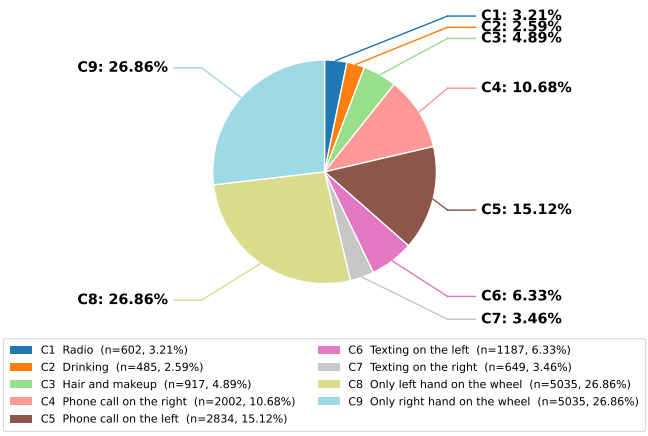
where γ is the wheel-class boost factor.

During sampling, an image is preferred if it increases the number of under-represented class instances and penalized if it only increases already saturated classes. The sampling score is defined as:

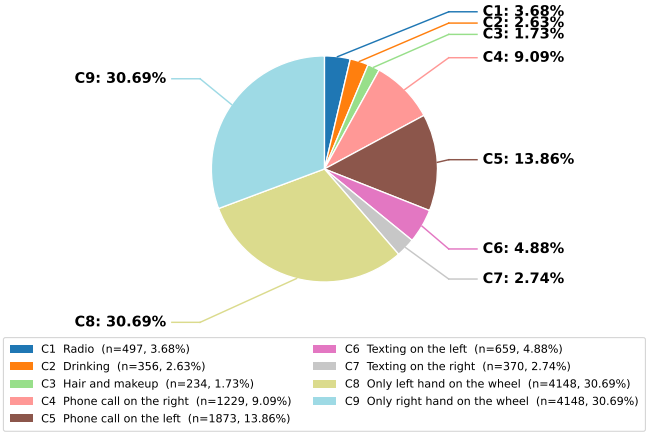
$$S(x) = \lambda_1 \sum_{c \in \mathcal{Y}(x)} \min\left(m_c(x), \max(0, T_c - h_c)\right) - \lambda_2 \sum_{c \in \mathcal{Y}(x)} \max\left(0, h_c + m_c(x) - T_c\right), \tag{3}$$

where $\mathcal{Y}(x)$ is the set of classes in image x, $m_c(x)$ is the number of instances of class c in x, h_c is the current sampled instance count for class c, and λ_1 and λ_2 control the reward for useful classes and the penalty for overflow classes. This process continues until the target class counts are reached or no sample provides positive gain.

This balanced sampling strategy keeps the final training set compact and prevents frequent classes from overwhelming rare distracted behaviors. Therefore, the final 9.3K RGB and 6.7K IR training frames are not simply the remaining valid frames after filtering, but a balanced subset selected from a much larger mined and verified candidate



(a) RGB training subset



(b) IR training subset

Fig. 1: Class distributions of the final pseudo-labeled training subsets for RGB and IR. The class-balanced sampling strategy reduces the dominance of frequent wheel-related classes while preserving sufficient samples for rare distracted behaviors.

pool. Figure 1 further visualizes the class distributions of the final balanced RGB and IR training subsets.

II. Evaluation Metrics

We evaluate DataDet from two perspectives: detection performance and deployment efficiency. Detection performance reflects whether the model can correctly recognize fine-grained distracted-driving behaviors, while deployment efficiency measures whether the model can run in real time on resource-constrained edge devices.

A. Detection Performance Metrics

For each distracted-driving class k, we compute true positives (TP_k), false positives (FP_k), false negatives (FN_k), and true negatives (TN_k). A true positive means that a sample of class k is correctly predicted as class k. A false positive means that a sample from another class is incorrectly predicted as class k. A false negative means that a sample of class k is incorrectly predicted as another class. A true negative means that a sample does not belong to class k and is not predicted as class k.

1) Precision

Precision measures the reliability of positive predictions for each class:

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}. \tag{4}$$

A higher precision means that when the model predicts a certain distracted-driving category, the prediction is more likely to be correct. In driver monitoring, this metric is

important because excessive false alarms may disturb the driver and reduce trust in the system.

2) Recall

Recall measures how many ground-truth samples of a class are successfully detected:

$$\mathrm{Recall}_k = \frac{TP_k}{TP_k + FN_k}. \tag{5}$$

A higher recall means that the model misses fewer samples of class k. This is important for safety-related scenarios, because missing risky behaviors such as phone calling or texting may reduce the practical effectiveness of the warning system.

3) F1-score

F1-score balances precision and recall:

$$\mathrm{F1}_k = \frac{2 \cdot \mathrm{Precision}_k \cdot \mathrm{Recall}_k}{\mathrm{Precision}_k + \mathrm{Recall}_k}. \tag{6}$$

F1-score becomes high only when both precision and recall are high. Therefore, it is suitable for evaluating fine-grained distracted-driving detection, where the model should avoid both false alarms and missed detections.

4) False Positive Rate

False positive rate (FPR) measures how often samples not belonging to class k are incorrectly predicted as class k:

$$\mathrm{FPR}_k = \frac{FP_k}{FP_k + TN_k}. \tag{7}$$

A lower FPR indicates fewer false triggers for a given class. This metric is especially important for real-time driver monitoring systems, since frequent false warnings may reduce system usability.

5) Macro-averaged Metrics

Because different distracted-driving categories may have different sample frequencies, we report macro-averaged metrics to give equal importance to each class. For K

evaluated classes, the macro-averaged metrics are computed as:

$$\mathrm{MacroPrecision} = \frac{1}{K}\sum_{k=1}^{K}\mathrm{Precision}_k, \tag{8}$$

$$\mathrm{MacroRecall} = \frac{1}{K}\sum_{k=1}^{K}\mathrm{Recall}_k, \tag{9}$$

$$\mathrm{MacroF1} = \frac{1}{K}\sum_{k=1}^{K}\mathrm{F1}_k, \tag{10}$$

$$\mathrm{MacroFPR} = \frac{1}{K}\sum_{k=1}^{K}\mathrm{FPR}_k\,. \tag{11}$$

Macro averaging prevents frequent classes from dominating the final result and better reflects the model's ability to recognize all distracted-driving categories.

B. Deployment Efficiency Metrics

In addition to detection accuracy, DataDet also targets real-time edge deployment. Therefore, we report model complexity and runtime efficiency metrics, including the number of parameters, FLOPs, inference latency, and FPS.

1) Number of Parameters

The number of parameters measures the storage and memory cost of a model:

$$\mathrm{Params} = \sum_{l=1}^{L}|\theta_l|, \tag{12}$$

where θ_l denotes the learnable parameters of layer l, and L is the total number of layers. We report this value in millions (M). A smaller number of parameters usually indicates lower memory usage and easier deployment on resource-constrained edge devices.

2) FLOPs

Floating-point operations (FLOPs) measure the computational cost of one forward pass. For convolutional layers, the FLOPs are mainly determined by the input/output feature size, kernel size, and channel numbers. We report FLOPs in billions (B). A lower FLOPs value generally indicates lower computational cost. This is important for embedded platforms, where the available computing resources and power budget are limited.

3) Inference Latency

Inference latency measures the time required by the model to process one image during inference:

$$\text{Latency} = t_{\text{infer}}, \tag{13}$$

where t_{infer} denotes the model inference time per image. In our implementation, latency is obtained from the official Ultralytics validation output using TensorRT engines, and only the inference time is used for reporting. The unit is milliseconds (ms). Lower latency means that the model can respond faster to incoming video frames, which is necessary for real-time distracted-driving detection.

4) Frames Per Second

Frames per second (FPS) measures the throughput of the model and is computed from inference latency:

$$\text{FPS} = \frac{1000}{\text{Latency}}, \tag{14}$$

where latency is measured in milliseconds. A higher FPS indicates that the model can process more frames per second. In edge-side driver monitoring, FPS directly reflects whether the detector can satisfy real-time video analysis requirements.

Overall, precision, recall, F1-score, and FPR evaluate the recognition quality of distracted-driving behaviors, while parameters, FLOPs, latency, and FPS evaluate the feasibility of real-time deployment. Together, these metrics provide a comprehensive assessment of both accuracy and efficiency.

III. Baseline

A. Baseline Selection

Most existing distracted-driving recognition methods are developed under a data-sufficient supervised setting. As summarized in recent surveys [1], [2], prior studies mainly focus on improving recognition accuracy through stronger visual, temporal, lightweight, sensor-based, or multimodal models. These methods usually assume that manually annotated training data are already available. Their evaluation therefore mainly measures the effectiveness of model architectures or feature representations under fixed labeled datasets.

DataDet addresses a different and more resource-constrained problem. It assumes the absence of manually annotated training data for the target domain. Instead, it aims to

convert continuous unlabeled driving videos into training data through edge-side event mining, cloud-side pseudo-labeling, label verification, and lightweight edge deployment. This setting is important in real deployment because continuous in-cabin videos are large, redundant, and expensive to annotate, while distracted behaviors are sparse and fine-grained. Therefore, the main challenge is not only how to design a stronger classifier, but also how to obtain reliable training data when labels, bandwidth, and edge computation are limited.

For this reason, directly comparing DataDet with prior supervised distracted-driver models under their original settings is not straightforward. Such a comparison would mix different tasks and supervision regimes. Prior methods evaluate recognition performance with human-labeled training data, whereas DataDet evaluates whether a deployable detector can be trained from automatically constructed pseudo labels. The performance gap in this case may reflect differences in data availability and annotation setting rather than the real contribution of either method.

Retraining prior distracted-driving models with DataDet-generated pseudo labels is also not the main focus of our evaluation. The downstream detector in DataDet is designed to be replaceable. If a previous model is retrained with our pseudo labels, it becomes another downstream detector inside the DataDet pipeline. This would not evaluate the original supervised design of that method. It would mainly test whether a different detector can consume the data constructed by DataDet.

Accordingly, we use two types of baselines. First, we compare with zero-shot VLM baselines. These models directly predict distracted-driving categories from test images without task-specific fine-tuning or manually labeled training data. This provides an annotation-free reference and examines whether direct large-model inference can solve the task. Second, we train multiple lightweight YOLO variants using the same DataDet-generated pseudo labels, including YOLOv5nu, YOLOv8n, YOLOv10n, and YOLO26n. Although they are all YOLO-family detectors, they use different network designs and detection heads. This comparison helps determine whether the observed performance is mainly due to the proposed data construction pipeline or to a specific downstream detector architecture.

Table III summarizes the difference between DataDet and common baseline settings. Compared with fully supervised distracted-driving methods, DataDet focuses on low-resource training data construction without manual target-domain labels. Compared with zero-shot VLM inference, DataDet uses large models only as offline cloud-side

annotation tools, while the final deployed model remains lightweight and edge-friendly.

Aspect	Supervised Methods	Zero-shot VLMs	DataDet
Task setting	Labeled-data recognition	Direct test-time recognition	Low-resource data construction
Supervision	Human labels	No training	Auto-generated pseudo labels
Large-model role	Not central	Direct inference	Cloud-side annotation only
Edge deployment	Model-dependent	Impractical	Lightweight detector on Jetson
Evaluation focus	Model accuracy	Zero-shot recognition	Data construction + edge detection

TABLE III: Comparison of baseline settings. DataDet focuses on low-resource data construction and lightweight edge deployment rather than fully supervised model comparison.

Overall, our baseline design follows the low-resource objective of DataDet. Zero-shot VLMs represent annotation-free large-model inference, while the YOLO variants represent replaceable lightweight edge detectors trained from the same automatically constructed data. Together, these baselines provide a more suitable evaluation for DataDet than directly comparing with supervised models trained on different manually labeled datasets.

IV. Implementation Details

A. Edge-side Event Mining Parameters

This section provides additional implementation details for the edge-side event mining module described in the main paper. The module uses the lightweight YOLO26n-seg model to segment the driver's left hand, right hand, and steering wheel. The predicted masks are then used to construct a stable steering wheel reference, estimate hand-wheel contact states, and extract candidate high-risk events from continuous video streams.

As described in the steering wheel stabilization module, each frame produces a steering wheel mask $M_{S,t}$ through the lightweight segmentation model M_θ. The stable steering wheel mask is obtained by accumulating the masks over the first W frames:

$$\tilde{M}_S = \frac{1}{W} \sum_{t=1}^{W} M_{S,t}. \tag{15}$$

In our implementation, the warm-up duration is set to 30 seconds. At least 15 valid steering wheel observations are required during this stage. A pixel is retained in the stabilized wheel mask if its occurrence frequency is no lower than 0.35 among the valid warm-up observations. The resulting mask is further processed by morphological closing and dilation. Steering wheel regions with an area smaller than 800 pixels are discarded. After the warm-up stage, the finalized steering wheel mask $\hat{M}_S$ is frozen and used as the

fixed spatial reference for the remaining frames. When the mask shape is incomplete, ellipse fitting is applied to obtain a more complete wheel-like representation.

After obtaining $\hat{M}_S$, DataDet estimates the hand-wheel state by comparing the left and right hand masks with the fixed steering wheel mask:

$$\mathcal{H}_t = f_{\text{hand-on-wheel}}(M_{L,t}, M_{R,t}, \hat{M}_S), \tag{16}$$

where $\mathcal{H}_t$ denotes the hand state at frame t. Before this comparison, small fragmented hand masks are removed. Specifically, hand masks with an area smaller than 1200 pixels are discarded. Multiple hand fragments are merged when their mask IoU is greater than 0.45 or their containment ratio is greater than 0.75. A hand is considered to be on the steering wheel if the raw overlap between the hand mask and $\hat{M}_S$ is at least 20 pixels, or if the overlap with the dilated wheel mask is at least 40 pixels. If no sufficient overlap exists, the minimum distance between the hand and wheel masks is used as an auxiliary criterion, and the hand is treated as on-wheel when the distance is no larger than 12 pixels.

For each frame, the hand state is represented as

$$\mathcal{H}_t = \{h_{L,t}, h_{R,t}\}, \quad h_{i,t} \in \{0, 1\}, \tag{17}$$

where $h_{i,t} = 1$ indicates that hand i is on the steering wheel, and $h_{i,t} = 0$ indicates that it is off the wheel. The frame-level risk score is then computed as:

$$R_t = 1 - \frac{\sum_{i \in \{L,R\}} h_{i,t}}{2}. \tag{18}$$

In our implementation, a frame is regarded as normal only when exactly two hands are detected on the wheel. Otherwise, it is treated as a high-risk frame. This setting follows the safety-oriented assumption that normal driving should maintain two-hand control, while missing, occluded, or off-wheel hands should be conservatively considered abnormal during event mining.

To reduce noise caused by occasional segmentation errors, DataDet applies temporal stabilization before starting or ending an event. A high-risk event is triggered only when the abnormal state lasts for 1 second, and it is terminated only when the normal state is continuously observed for 1 second. This hysteresis strategy prevents short-term mask jitter or transient hand occlusion from producing unstable event boundaries.

Equivalently, in the sliding-window formulation of the main paper,

$$\bar{R}_t = \frac{1}{W} \sum_{k=t-W+1}^{t} R_k, \tag{19}$$

the implementation uses a one-second temporal persistence constraint for both event activation and event termination.

When a candidate event is detected, the corresponding continuous frames are marked as:

$$\mathcal{E}_c = \{I_k \,|\, t - W + 1 \le k \le t, \bar{R}_t \ge \tau\}. \tag{20}$$

To reduce redundant uploads, DataDet does not save every frame in $\mathcal{E}_c$. Instead, event frames are sampled every 5 frames. Each event folder contains at most 15 sampled frames. If an event exceeds this limit, it is split into multiple event parts. For each saved frame, the abnormal hand masks are converted into bounding boxes and stored in normalized YOLO format. These sampled event frames and hand boxes are then used as the input to the cloud-side annotation and verification module.

B. RTMPose-based Hand Box Tool

This section provides the implementation details of the hand bounding-box tool used in the cloud-side annotation stage. In the main paper, the hand box generation process is defined as follows. Given an event frame set

$$\mathcal{F} = \{I_1, I_2, \ldots, I_t\}, \tag{21}$$

RTMPose-WholeBody detects full-body keypoints for each frame. The left-hand and right-hand keypoint sets are denoted as:

$$\mathcal{K}_{\mathrm{L},t} = \{(x_i, y_i, s_i)\}_{i=1}^{21}, \quad \mathcal{K}_{\mathrm{R},t} = \{(x_j, y_j, s_j)\}_{j=1}^{21}, \tag{22}$$

where (x, y) denotes the image coordinate and s denotes the keypoint confidence score.

In our implementation, we use RTMPose-WholeBody through `rtmlib`. The model is initialized with the COCO-WholeBody layout, which contains 133 keypoints for each detected person. The 133 keypoints consist of 17 body keypoints, 6 foot keypoints, 68 face keypoints, and 42 hand keypoints. The hand keypoints are further divided into 21 left-hand keypoints and 21 right-hand keypoints. Specifically, indices 91–111 correspond to the left hand, and indices 112–132 correspond to the right hand.

For each event frame, if multiple persons are detected, we select the person with the largest accumulated hand-keypoint confidence. This is consistent with the in-cabin

driver-monitoring setting, where the target subject is the driver. For each hand, only keypoints with confidence scores no lower than s_{thr} are used:

$$\mathcal{K}_{i,t}^{+} = \{(x, y, s) \in \mathcal{K}_{i,t} \mid s \geq s_{\text{thr}}\}, \quad i \in \{\text{L}, \text{R}\}. \tag{23}$$

We set $s_{\text{thr}} = 0.2$ in our experiments. The hand bounding box is then computed as the minimum enclosing rectangle of the valid hand keypoints:

$$x_{\min} = \min_{(x,y,s)\in\mathcal{K}_{i,t}^{+}} x, \quad y_{\min} = \min_{(x,y,s)\in\mathcal{K}_{i,t}^{+}} y, \tag{24}$$

$$x_{\max} = \max_{(x,y,s)\in\mathcal{K}_{i,t}^{+}} x, \quad y_{\max} = \max_{(x,y,s)\in\mathcal{K}_{i,t}^{+}} y. \tag{25}$$

The box is clipped to the image boundary and converted into the normalized YOLO format:

$$\mathcal{B}_{i,t} = (c_x, c_y, w, h), \tag{26}$$

where

$$c_x = \frac{x_{\min} + x_{\max}}{2W}, \quad c_y = \frac{y_{\min} + y_{\max}}{2H}, \tag{27}$$

$$w = \frac{x_{\max} - x_{\min}}{W}, \quad h = \frac{y_{\max} - y_{\min}}{H}. \tag{28}$$

Here, W and H denote the image width and height.

The generated label file contains only the normalized hand boxes. The first line stores the left-hand box, and the second line stores the right-hand box:

$$\texttt{line}_1 = \mathcal{B}_{\text{L},t}, \quad \texttt{line}_2 = \mathcal{B}_{\text{R},t}. \tag{29}$$

No class ID is written at this stage, because the semantic hand behavior labels are generated later by the cloud-side agent. If either the left hand or the right hand cannot be reliably localized, the corresponding label file is left empty. Such frames are not forced into the training set, which avoids introducing low-quality hand boxes into subsequent pseudo-label generation and detector training.

In our experiments, RTMPose-WholeBody is executed on GPU with the ONNXRuntime backend and the performance mode. This tool is only used in the cloud-side data construction stage and is not deployed on the edge device during real-time inference.

C. Agent Prompt Design

To make cloud-side pseudo-labeling controllable and reproducible, DataDet does not use one open-ended VLM prompt to annotate the whole image. Instead, it divides the annotation process into three stages: left-hand labeling, right-hand labeling, and label

verification. This design reduces the ambiguity of two-hand reasoning. It also makes the VLM rely on visible hand regions, object evidence, temporal context, and physical constraints.

The first two prompts are single-hand annotators. As shown in Tables V and VI, the left-hand annotator only focuses on the green "L" hand box. The right-hand annotator only focuses on the yellow "R" hand box. Each annotator first checks the raw image evidence. It then converts visible cues into symbolic evidence states. After that, it resolves one label using a fixed priority order. Consecutive frames with the same label are finally merged into segments. This stepwise process avoids direct labeling from a rough visual impression. It also reduces confusion among similar behaviors, such as phone calling, texting, face touching, radio operation, and wheel holding.

The prompts also use strict visibility and physical-consistency rules. Phone-related labels require visible phone evidence or strong continuity from nearby frames. Drinking requires a visible drink container near the mouth. Radio requires visible console interaction. Wheel holding requires visible contact with the steering wheel. In addition, left-hand labels cannot use right-hand-specific classes. Right-hand labels also cannot use left-hand-specific classes. Frames with severe occlusion, blur, missing hands, or uncertain evidence are skipped. They are not forced into noisy labels.

After the two single-hand annotation stages, DataDet applies a verifier prompt, as shown in Table VII. The verifier takes the raw images and the two JSON outputs as inputs. It is not asked to relabel all frames from scratch. By default, it keeps the previous labels. It changes a label only when the raw image shows clear contradictory evidence. This conservative strategy reduces unnecessary label drift. It can also correct obvious errors, especially for ambiguous right-hand states such as weak wheel holding or generic off-wheel labels.

Finally, the verifier regroups the corrected frame-level labels into segment-level pseudo-labels. It returns a strict JSON object. Only verified and non-skipped frames are used to construct training samples. In this way, DataDet turns VLM-based visual reasoning into a constrained and auditable pseudo-labeling process. This improves the reliability of automatically generated training data.

D. Edge Devices and Runtime Environment

We evaluate DataDet on real embedded devices to verify its edge deployment feasibility. The edge testbed includes two NVIDIA Jetson AGX Orin Developer Kits

Item	Jetson AGX Orin	Jetson Xavier NX
Number of devices	2	3
Device type	Developer Kit	Developer Kit
Module	NVIDIA Jetson AGX Orin	NVIDIA Jetson Xavier NX
Memory	64 GB RAM	8 GB RAM
SoC	tegra23x	tegra194
CUDA architecture	8.7	7.2
Operating system	Ubuntu 20.04	Ubuntu 20.04
Kernel version	5.10.216-tegra	5.10.216-tegra
Python	3.8.10	3.8.10
CUDA	11.4.315	11.4.315
cuDNN	8.6.0.166	8.6.0.166
TensorRT	8.5.2.2	8.5.2.2
VPI	2.4.8	2.4.8
Vulkan	1.3.204	1.3.204
OpenCV	4.8.0 with CUDA	4.10.0 with CUDA
JetPack / L4T	JetPack 5.1.5 / L4T 35.6.3	JetPack 5.1.5 / L4T 35.6.3

TABLE IV: Edge devices and runtime environment used for deployment evaluation.

and three NVIDIA Jetson Xavier NX Developer Kits. The two platforms represent different edge-computing budgets. AGX Orin is used as a higher-performance edge device, while Xavier NX represents a more resource-constrained setting.



Fig. 2: Jetson-based edge testbed used in our experiments. It contains two Jetson AGX Orin Developer Kits and three Jetson Xavier NX Developer Kits.

Table IV summarizes the hardware and software environment. Both platforms run Ubuntu 20.04 with JetPack 5.1.5 / L4T 35.6.3. The inference environment is built with

Role	Prompt Content
System	You are an intelligent in-cabin vision assistant for per-frame **left-hand** driver behavior labeling. You must strictly follow the user-provided label set, visibility rules, physical-consistency constraints, temporal-consistency rules, and output requirements. Return **only** the specified JSON object with no extra text.

User	**Task:** You will be given exactly {N} side-profile images of a driver, named image_1 to image_{N} in chronological order. Each image may contain hand boxes: Green box labeled **"L" = LEFT hand**; Yellow box labeled **"R" = RIGHT hand**. Use the raw images as the only evidence source. Use the green **"L"** box only as the localization cue for the **left hand**. This stage predicts only the **left-hand label** for each usable frame and then merges consecutive frames with the same left-hand label.
	
	Allowed left-hand labels: For each usable frame, assign exactly one label from: radio, drinking, hair_and_makeup, phonecall_left, texting_left, hold_on_the_wheel_left, not_hold_on_the_wheel. Do **not** output right-hand labels or head labels.
	
	Reasoning process: Internally follow four steps: (1) inspect raw image evidence, including left-hand location, contact, and visible objects; (2) convert visual cues into symbolic left-hand evidence states; (3) resolve one left-hand label according to the mapping rules; and (4) merge consecutive frames with identical left-hand labels. Do **not** jump directly from rough impressions to final labels.
	
	Evidence priority: Use the current frame as the primary evidence. Nearby frames image_{i-1} and image_{i+1} may only stabilize ambiguous cases, and must not override clear contradictory evidence in the current frame. Do not invent invisible objects or unsupported interactions.
	
	Symbolic evidence states: For each readable frame, internally determine the following left-hand evidence states: L_PHONE_AT_EAR_VISIBLE, L_PHONE_FRAGMENT_VISIBLE, L_PHONECALL_CONTINUITY, L_PHONECALL_EVIDENCE; L_PHONE_LOW_VISIBLE, L_TEXTING_POSTURE, L_TEXTING_EVIDENCE; L_DRINK_VISIBLE, L_DRINK_TO_MOUTH, L_DRINKING_EVIDENCE; L_FACE_TOUCH_VISIBLE, L_GROOMING_OBJECT_VISIBLE, L_FACE_TOUCH_EVIDENCE; L_CONSOLE_VISIBLE, L_CONSOLE_DIRECTED_POSE, L_CONSOLE_INTERACTION_VISIBLE, L_CONSOLE_CONTINUITY, L_RADIO_EVIDENCE; L_WHEEL_VISIBLE_AT_CONTACT, L_WHEEL_CONTACT_VISIBLE, L_WHEEL_HOLD_POSE, L_WHEEL_HOLD_EVIDENCE; L_CLEAR_NON_WHEEL_REGION, L_NOT_IN_STABLE_WHEEL_POSE, and L_OFF_WHEEL_EVIDENCE. These states are only for internal reasoning and must not be output.
	
	Observation rules: phonecall_left requires visible phone evidence at or near the left ear/upper-cheek region, a distinct phone fragment, or strong temporal continuity supported by adjacent frames. Near-ear pose alone is not enough. texting_left requires a visible phone below head level and a compatible phone-use posture. drinking requires a visible drink container held by the left hand and contacting or being brought to the mouth. hair_and_makeup requires visible face/hair contact or a visible grooming object; near-face pose without contact is not enough. radio requires a visible console region and a console-directed left-hand pose, with either visible interaction or stable console-reaching continuity. hold_on_the_wheel_left requires visible wheel contact at the left-hand contact region and a stable wheel-hold pose; proximity alone is not enough. not_hold_on_the_wheel is used only when no stronger action evidence exists and the left hand is clearly off the wheel or not in a stable wheel-hold pose.
	
	Label resolution priority: Use the following strict priority: phonecall_left > texting_left > drinking > hair_and_makeup > radio > hold_on_the_wheel_left > not_hold_on_the_wheel. If phone evidence exists, prefer phonecall_left over hair_and_makeup. If visible face-touch is absent, do not use hair_and_makeup. If console interaction is stable, prefer radio over generic off-wheel. If no reliable label can be assigned, skip the frame.
	
	Usable frame rule: Skip a frame if the left hand box exists but the actual left hand is not visible, or if the left hand is too occluded, blurred, cut off, or visually ambiguous to assign one reliable label. Skipped frames must not appear in the output.
	
	Hand-box sanity check: If any left or right hand box is abnormally large, such as covering more than about one third of the image width or height or extending far beyond the real hand/arm area, set "hand_box_status":"abnormal_large". Otherwise set "hand_box_status":"normal". This field must be output even though this stage predicts only left-hand labels.
	
	Segment grouping: Group consecutive non-skipped frames into the same segment only when their left_hand_label is identical. Frame indices must be sorted in ascending order.
	
	Output: Return only a JSON object with exactly two keys: "segments" and "hand_box_status". Each segment object must contain exactly "frames" and "left_hand_label". No extra keys, comments, or explanations are allowed.
	
	`{`
	`  "segments": [`
	`    {`
	`      "frames": [<int>, ...],`
	`      "left_hand_label": "<class>"`
	`    }`
	`  ],`
	`  "hand_box_status": "<normal|abnormal_large>"`
	`}`

TABLE V: Prompt template used for the left-hand behavior annotator. The prompt constrains the VLM to perform evidence-based left-hand labeling, enforce object-visibility and temporal-consistency rules, and return a JSON-only segment-level output.

CUDA 11.4.315, cuDNN 8.6.0.166, and TensorRT 8.5.2.2. OpenCV with CUDA support is used for video decoding, frame preprocessing, and runtime integration. Keeping the software stack consistent across platforms makes the latency and FPS comparison more controlled.

In DataDet, the Jetson devices are used in two stages. During data construction, they run the lightweight event mining module and select high-risk candidate clips from continuous in-cabin videos. During deployment, the trained lightweight detector is converted to TensorRT and executed locally on the same devices for real-time distracted-driving detection. This setup allows us to measure both front-end event mining efficiency and final edge inference performance under realistic embedded hardware constraints.

V. Additional Experimental Analysis

A. Interaction between Annotation Models and Edge Detectors

Table VIII analyzes how different annotation models, inference modes, and edge detectors jointly affect the final distracted driving detection performance. We first compare DataDet with direct zero-shot VLM inference. Although stronger VLMs can

Role	Prompt Content
System	You are an intelligent in-cabin vision assistant for per-frame **right-hand** driver behavior labeling. You must strictly follow the user-provided label set, visibility rules, physical-consistency constraints, temporal-consistency rules, and output requirements. Return **only** the specified JSON object with no extra text.

User	**Task:** You will be given exactly {N} side-profile images of a driver, named image_1 to image_{N} in chronological order. Each image may contain hand boxes: Green box labeled **"L" = LEFT hand**; Yellow box labeled **"R" = RIGHT hand**. Use the raw images as the only evidence source. Use the yellow **"R"** box only as the localization cue for the **right hand**. This stage predicts only the **right-hand label** for each usable frame and then merges consecutive frames with the same right-hand label.
	
	Allowed right-hand labels: For each usable frame, assign exactly one label from: radio, drinking, hair_and_makeup, phonecall_right, texting_right, hold_on_the_wheel_right, not_hold_on_the_wheel. Do **not** output left-hand labels or head labels.
	
	Reasoning process: Internally follow four steps: (1) inspect raw image evidence, including right-hand location, contact, and visible objects; (2) convert visual cues into symbolic right-hand evidence states; (3) resolve one right-hand label according to the mapping rules; and (4) merge consecutive frames with identical right-hand labels. Do **not** jump directly from rough impressions to final labels.
	
	Evidence priority: Use the current frame as the primary evidence. Nearby frames image_{i-1} and image_{i+1} may only stabilize ambiguous cases, and must not override clear contradictory evidence in the current frame. Do not invent invisible objects or unsupported interactions.
	
	Symbolic evidence states: For each readable frame, internally determine the following right-hand evidence states: R_PHONE_AT_EAR_VISIBLE, R_PHONECALL_CONTINUITY, R_PHONECALL_EVIDENCE; R_PHONE_LOW_VISIBLE, R_TEXTING_POSTURE, R_TEXTING_CONTINUITY, R_TEXTING_EVIDENCE; R_DRINK_VISIBLE, R_DRINK_TO_MOUTH, R_DRINKING_EVIDENCE; R_FACE_TOUCH_VISIBLE, R_GROOMING_OBJECT_VISIBLE, R_FACE_TOUCH_EVIDENCE; R_CONSOLE_VISIBLE, R_CONSOLE_DIRECTED_POSE, R_CONSOLE_INTERACTION_VISIBLE, R_CONSOLE_CONTINUITY, R_RADIO_EVIDENCE; R_WHEEL_VISIBLE_AT_CONTACT, R_WHEEL_CONTACT_VISIBLE, R_WHEEL_HOLD_POSE, R_WHEEL_HOLD_EVIDENCE; R_CLEAR_NON_WHEEL_REGION, R_NOT_IN_STABLE_WHEEL_POSE, and R_OFF_WHEEL_EVIDENCE. These states are only for internal reasoning and must not be output.
	
	Observation rules: phonecall_right requires visible phone evidence at or near the right ear/cheek region, or strong temporal continuity supported by adjacent frames. Near-ear pose alone is not enough. texting_right requires a visible phone below head level with a compatible phone-use posture, or strong low-hand phone-use continuity without contradictory evidence. drinking requires a visible drink container held by the right hand and contacting or being brought to the mouth. hair_and_makeup requires visible face/hair contact or a visible grooming object; near-face pose without contact is not enough. radio requires a visible console region and a console-directed right-hand pose, with either visible interaction or stable console-reaching continuity. hold_on_the_wheel_right requires visible wheel contact at the right-hand contact region and a stable wheel-hold pose; proximity alone is not enough. not_hold_on_the_wheel is used only when no stronger action evidence exists and the right hand is clearly off the wheel or not in a stable wheel-hold pose.
	
	Label resolution priority: Use the following strict priority: phonecall_right > texting_right > drinking > hair_and_makeup > radio > hold_on_the_wheel_right > not_hold_on_the_wheel. If phone evidence exists, prefer phonecall_right over hair_and_makeup. If visible face-touch is absent, do not use hair_and_makeup. If console interaction is stable, prefer radio over generic off-wheel. If no reliable label can be assigned, skip the frame.
	
	Usable frame rule: Skip a frame if the right hand box exists but the actual right hand is not visible, or if the right hand is too occluded, blurred, cut off, or visually ambiguous to assign one reliable label. Skipped frames must not appear in the output.
	
	Hand-box sanity check: If any left or right hand box is abnormally large, such as covering more than about one third of the image width or height or extending far beyond the real hand/arm area, set "hand_box_status":"abnormal_large". Otherwise set "hand_box_status":"normal". This field must be output even though this stage predicts only right-hand labels.
	
	Segment grouping: Group consecutive non-skipped frames into the same segment only when their right_hand_label is identical. Frame indices must be sorted in ascending order.
	
	Output: Return only a JSON object with exactly two keys: "segments" and "hand_box_status". Each segment object must contain exactly "frames" and "right_hand_label". No extra keys, comments, or explanations are allowed.
	
	`{`
	`  "segments": [`
	`    {`
	`      "frames": [<int>, ...],`
	`      "right_hand_label": "<class>"`
	`    }`
	`  ],`
	`  "hand_box_status": "<normal|abnormal_large>"`
	`}`

TABLE VI: Prompt template used for the right-hand behavior annotator. The prompt constrains the VLM to perform evidence-based right-hand labeling, enforce object-visibility and temporal-consistency rules, and return a JSON-only segment-level output.

provide reasonable recognition ability without training, their zero-shot performance remains limited on DMD. The best zero-shot setting achieves only 61.0% RGB F1 and 60.0% IR F1, with relatively high false positive rates. This indicates that directly using VLMs as online classifiers is insufficient for reliable edge-side deployment, especially under fine-grained hand–object and hand–wheel interactions. In contrast, once the VLM outputs are converted into pseudo labels and used to train lightweight edge detectors, DataDet consistently improves both accuracy and stability. For example, even with Gemma-4-26B-A4B in standard mode, DataDet improves RGB F1 from 45.7% to 87.8–89.8%, and IR F1 from 42.9% to 57.1–71.8%, depending on the edge detector scale.

The results further show that pseudo-label quality is a key factor, but its effect is not isolated from the downstream detector. In general, reasoning mode improves label accuracy and leads to better edge performance. For Gemma-4-26B-A4B, reasoning increases label accuracy from 70.2% to 90.2% on RGB and from 67.3% to 81.1% on IR, which raises the best RGB F1 from 89.8% to 94.2% and the best IR F1 from 71.8% to 78.1%. Similar trends are observed for Qwen-series models. Qwen3.5-35B-A3B

Role	Prompt Content
System	You are an intelligent in-cabin vision assistant for verifying driver hand-action labels. This is a **verification task**, not a full relabeling task. You must check whether the candidate left-hand and right-hand labels are supported by the raw images, correct them only when clear contradictory evidence exists, and return **only** the specified JSON object with no extra text.

User	**Task:** You will be given exactly {N} side-profile images of a driver, named `image_1` to `image_{N}` in chronological order. Each image may contain hand boxes: Green box labeled **"L" = LEFT hand**; Yellow box labeled **"R" = RIGHT hand**. Use these boxes only as localization and left/right identity cues.
	You will also be given two auxiliary JSON inputs: **Stage 1 left-hand result:** {{`STAGE1_LEFT_JSON`}}. **Stage 2 right-hand result:** {{`STAGE2_RIGHT_JSON`}}. Your job is to verify whether the stage results are supported by the raw images and correct them only when necessary.
	Core rule: Default behavior is to keep the stage1/stage2 label. Change a label only when the current frame provides **clear contradictory evidence** and a better valid label is clearly supported. Do not replace an existing label with a weaker guess. For each usable frame, first inspect the raw image, then check whether each candidate label is `supported` or `contradicted`. Raw image evidence has priority over stage results. Auxiliary inputs may only stabilize nearby continuity and must not replace missing object evidence.
	Allowed hand labels: `radio`, `drinking`, `hair_and_makeup`, `phonecall_right`, `phonecall_left`, `texting_left`, `texting_right`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `not_hold_on_the_wheel`. Hard constraints: LEFT must never be `phonecall_right`, `texting_right`, or `hold_on_the_wheel_right`; RIGHT must never be `phonecall_left`, `texting_left`, or `hold_on_the_wheel_left`.
	Usable frame rule: Skip the frame if a hand box is present but the corresponding hand is not visible, either hand is too blurred, occluded, or cropped to label confidently, or both hand labels cannot be confidently assigned. Skipped frames must not appear in the output.
	Internal verification states: For each usable frame, internally infer the following states only. For the left hand: `V_LEFT_PHONECALL`, `V_LEFT_TEXTING`, `V_LEFT_DRINKING`, `V_LEFT_FACE_TOUCH`, `V_LEFT_RADIO`, `V_LEFT_WHEEL`, `V_LEFT_IDLE_OFF_WHEEL`. For the right hand: `V_RIGHT_PHONECALL`, `V_RIGHT_TEXTING`, `V_RIGHT_DRINKING`, `V_RIGHT_FACE_TOUCH`, `V_RIGHT_RADIO`, `V_RIGHT_WHEEL_STRONG`, `V_RIGHT_IDLE_OFF_WHEEL`. Also infer `V_STAGE1_LEFT_OK` and `V_STAGE2_RIGHT_OK`. These states must not be output.
	Left-hand evidence rules: Set `V_LEFT_PHONECALL` only with visible phone at the left ear/cheek or clear left-side phone continuity. Set `V_LEFT_TEXTING` only with visible phone below head level. Set `V_LEFT_DRINKING` only with a visible drink contacting or approaching the mouth. Set `V_LEFT_FACE_TOUCH` only with visible face/hair contact or a visible grooming object. Set `V_LEFT_RADIO` only with visible console and clear console-reaching or interaction. Set `V_LEFT_WHEEL` only with visible wheel and visible left-hand contact. Set `V_LEFT_IDLE_OFF_WHEEL` only when no stronger left-hand state is true and the hand is clearly not wheel-holding. Since the left hand is already predicted in stage1, be conservative and change left-hand labels only when contradiction is clear.
	Right-hand evidence rules: Set `V_RIGHT_PHONECALL` only with visible phone at the right ear/cheek or clear right-side phone continuity. Set `V_RIGHT_TEXTING` only with visible phone below head level or very clear low-phone continuity. Set `V_RIGHT_DRINKING` only when a visible drink container exists and is touching, immediately in front of, or clearly approaching the mouth. Set `V_RIGHT_FACE_TOUCH` only with visible face/hair contact, clear pressing/rubbing/adjusting of face or hair, or a visible grooming object. Set `V_RIGHT_RADIO` only with visible console interaction, or a visible console region and a stable control-like reaching pose. Set `V_RIGHT_WHEEL_STRONG` only when the wheel is visible at the contact region, visible right-hand contact exists, the pose is a stable wheel-hold pose, and no stronger specific action explains the frame better. Set `V_RIGHT_IDLE_OFF_WHEEL` only when no stronger right-hand state is true, no stable wheel-hold is visible, and the hand is in a clear idle off-wheel position.
	Special challenger rule for the right hand: The main correction targets are stage2 frames labeled `hold_on_the_wheel_right` or `not_hold_on_the_wheel`. If stage2 says `hold_on_the_wheel_right`, do not keep it unless `V_RIGHT_WHEEL_STRONG = true`. Replace it if `V_RIGHT_PHONECALL`, `V_RIGHT_TEXTING`, `V_RIGHT_DRINKING`, `V_RIGHT_FACE_TOUCH`, or `V_RIGHT_RADIO` is clearly supported. Wheel nearby or weak overlap is not enough. If stage2 says `not_hold_on_the_wheel`, do not keep it unless `V_RIGHT_IDLE_OFF_WHEEL = true`. Replace it if a stronger visible action or strong wheel-hold evidence is supported.
	Stage support check: Set `V_STAGE1_LEFT_OK = true` if the stage1 left label matches current evidence or is acceptable under nearby continuity and is not contradicted. Set `V_STAGE2_RIGHT_OK = true` only if the stage2 right-hand label matches the strongest supported current evidence: `phonecall_right` requires `V_RIGHT_PHONECALL`; `texting_right` requires `V_RIGHT_TEXTING`; `drinking` requires `V_RIGHT_DRINKING`; `hair_and_makeup` requires `V_RIGHT_FACE_TOUCH`; `radio` requires `V_RIGHT_RADIO`; `hold_on_the_wheel_right` requires `V_RIGHT_WHEEL_STRONG`; and `not_hold_on_the_wheel` requires `V_RIGHT_IDLE_OFF_WHEEL`. If contradicted by stronger visible evidence, set it to false.
	Final correction rule: If `V_STAGE1_LEFT_OK = true`, keep the stage1 left label; otherwise assign the first supported label in this order: `phonecall_left` > `texting_left` > `drinking` > `hair_and_makeup` > `radio` > `hold_on_the_wheel_left` > `not_hold_on_the_wheel`. For the right hand, if `V_STAGE2_RIGHT_OK = true`, keep the stage2 right label; otherwise assign the first supported label in this order: `phonecall_right` > `texting_right` > `drinking` > `hair_and_makeup` > `radio` > `hold_on_the_wheel_right` > `not_hold_on_the_wheel`. Specific visible right-hand actions beat weak wheel-hold. Do not replace a right-hand label with off-wheel unless off-wheel is clearly visible.
	Temporal consistency: Use nearby frames only to stabilize phone continuity, texting continuity, console-reaching continuity, and wheel-hold continuity. Do not let continuity override clear contradictory evidence in the current frame.
	Hand-box sanity check: If any left or right hand box is abnormally large, such as covering more than about one third of the image width or height or extending far beyond the real hand/arm area, set `"hand_box_status":"abnormal_large"`. Otherwise set `"hand_box_status":"normal"`.
	Segment regrouping: After verification and correction, regroup consecutive non-skipped frames only when both `left_hand_label` and `right_hand_label` are identical. Frame indices must be sorted in ascending order.
	Output: Return only a JSON object with exactly two keys: `"segments"` and `"hand_box_status"`. Each segment object must contain exactly `"frames"`, `"left_hand_label"`, and `"right_hand_label"`. Skipped frames must not appear. No extra keys, comments, or explanations are allowed.
	`{`
	`  "segments": [`
	`    {`
	`      "frames": [<int>, ...],`
	`      "left_hand_label": "<class>",`
	`      "right_hand_label": "<class>"`
	`    }`
	`  ],`
	`  "hand_box_status": "<normal|abnormal_large>"`
	`}`

TABLE VII: Prompt template used for the verifier. The verifier checks the left- and right-hand labels generated by the previous stages, preserves supported labels, corrects only clearly contradicted labels, and outputs verified segment-level pseudo-labels in a strict JSON format.

with reasoning obtains the highest pseudo-label accuracy, 97.2% on RGB and 95.9% on IR, and also achieves the best IR detection result, with YOLO26x reaching 87.6% F1 and only 0.9% FPR. Qwen3.6-35B-A3B with reasoning obtains slightly lower label accuracy, 96.8% on RGB and 95.7% on IR, but produces the best RGB result with YOLO26n, reaching 94.6% F1 and 0.5% FPR. These results suggest that higher label accuracy generally benefits the edge detector, while the final performance also depends on how well the pseudo-label distribution matches the detector capacity and modality.

The interaction between annotation quality and detector scale is also evident. When pseudo labels are relatively noisy, increasing the detector size does not always bring consistent gains. For example, under Ovis2.6-30B-A3B standard mode, RGB label accuracy is only 80.3%, and the RGB F1 of YOLO26n–YOLO26x ranges from 81.0%

to 85.8%, showing limited improvement despite larger models. By contrast, when high-quality reasoning-based pseudo labels are used, larger detectors usually provide stronger IR performance. With Qwen3.5-35B-A3B reasoning, IR F1 increases from 81.5% with YOLO26n to 87.6% with YOLO26x, while the IR FPR decreases from 1.6% to 0.9%. This indicates that larger edge detectors can better exploit high-quality pseudo labels, especially in the more challenging IR modality.

However, the best detector is not always the largest one. In the RGB setting, YOLO26n trained with Qwen3.6-35B-A3B reasoning achieves the best F1-score of 94.6%, slightly outperforming larger variants. This suggests that RGB cues are sufficiently discriminative for many target behaviors, and a compact detector can already capture the relevant patterns when trained with accurate pseudo labels. In contrast, IR images contain weaker texture and color cues, making detector capacity more important. The best IR result is obtained by YOLO26x with Qwen3.5-35B-A3B reasoning, reaching 87.6% F1. Therefore, the choice of edge detector should consider both the pseudo-label quality and the sensing modality: compact models are effective for RGB deployment, while larger models provide stronger robustness for IR.

The per-class results in Tables IX–XVIII further support this observation. In RGB, high-quality reasoning-based annotations produce consistently strong per-class performance across most categories, especially for visually distinctive classes such as phone calling, radio interaction, and hand-on-wheel states. The remaining errors are concentrated in visually ambiguous categories such as hair and makeup or texting, where hand position, occlusion, and object visibility can vary substantially. In IR, the performance gap across classes becomes larger, and classes requiring small-object or fine-grained appearance cues are more sensitive to annotation model quality and detector scale. This explains why IR benefits more from stronger pseudo labels and larger edge detectors.

Overall, these results show that DataDet is not tied to a specific VLM or a specific YOLO variant. Instead, the annotation model and edge detector play complementary roles. Stronger reasoning-based annotation models improve pseudo-label quality, while appropriate edge detectors convert these labels into efficient deployment models. The best configuration depends on the target modality and deployment budget, but DataDet consistently bridges the gap between high-cost VLM reasoning and low-cost edge inference.

B. Impact of Reasoning Mode on Downstream Detector Training

Table VIII further examines whether reasoning mode only improves the annotation model itself, or whether such improvement can be transferred to the downstream edge detector through pseudo-label training. The results show that reasoning mode generally improves both pseudo-label accuracy and the final detector performance, indicating that more reliable cloud-side reasoning can be effectively converted into stronger edge-side models.

For zero-shot VLM inference, reasoning mode improves the recognition ability of most annotation models, but the overall performance remains limited. For example, Gemma-4-26B-A4B improves from 45.7% to 57.5% RGB F1 and from 42.9% to 51.1% IR F1 when reasoning is enabled. Qwen3.5-35B-A3B also improves from 54.6% to 61.0% RGB F1 and from 46.2% to 60.0% IR F1. However, even the best zero-shot reasoning result is still far below the trained DataDet detectors. This suggests that reasoning alone is not sufficient for real-time distracted driving detection, but it can provide higher-quality supervision for training lightweight edge models.

The effect of reasoning becomes more significant after the generated labels are used for detector training. For Gemma-4-26B-A4B, reasoning increases the pseudo-label accuracy from 70.2% to 90.2% on RGB and from 67.3% to 81.1% on IR. This directly improves the best RGB F1 from 89.8% to 94.2% and the best IR F1 from 71.8% to 78.1%. A similar trend is observed for Ovis2.6-30B-A3B. Although its IR label accuracy changes only slightly from 86.9% to 87.0%, reasoning substantially improves RGB label accuracy from 80.3% to 92.3%, leading to a large RGB F1 improvement from 85.8% to 92.6%. These results indicate that reasoning mode is especially useful when the standard inference mode produces noisy or incomplete pseudo labels.

The Qwen-series models show that high-quality reasoning labels can produce strong and stable downstream detectors. Qwen3.5-35B-A3B with reasoning achieves the highest pseudo-label accuracy, 97.2% on RGB and 95.9% on IR, and obtains the best IR performance with YOLO26x, reaching 87.6% F1 and 0.9% FPR. Qwen3.6-35B-A3B with reasoning achieves slightly lower label accuracy, 96.8% on RGB and 95.7% on IR, but produces the best RGB result with YOLO26n, reaching 94.6% F1 and 0.5% FPR. Compared with the corresponding standard mode, Qwen3.6-35B-A3B reasoning improves the best RGB F1 from 90.2% to 94.6% and the best IR F1 from 76.9% to 85.7%. This confirms that reasoning-enhanced pseudo labels provide more effective

supervision for edge detector training.

The improvement brought by reasoning mode is also reflected in false positive rates. For Gemma-4-26B-A4B, the best RGB FPR decreases from 1.0% under standard mode to 0.5% under reasoning mode, while the best IR FPR decreases from 2.9% to 2.1%. For Qwen3.5-35B-A3B, reasoning reduces the best IR FPR from 1.7% to 0.9%. These reductions suggest that reasoning does not merely increase recall by producing more positive predictions; instead, it helps generate cleaner pseudo labels that reduce confusion between distracted and non-distracted states during detector training.

Nevertheless, reasoning mode does not always guarantee uniform improvement for every model and modality. Holo3-35B-A3B improves its label accuracy from 85.3% to 89.5% on RGB and from 77.2% to 84.6% on IR, but its downstream detector performance does not consistently improve over the standard mode. For example, the best RGB F1 decreases from 91.3% to 90.3%, and the best IR F1 decreases from 75.0% to 71.7%. This indicates that label accuracy alone cannot fully explain downstream performance. The distribution of errors, class imbalance, and the consistency of pseudo labels across similar frames also affect how well the edge detector can learn from the generated annotations.

The per-class results provide further evidence for this observation. In RGB, reasoning-based Qwen models improve most fine-grained behavior categories and reduce average FPR to 0.5–0.7% across YOLO26 variants. The strongest gains appear in visually ambiguous categories, such as hair and makeup or texting-related actions, where step-by-step reasoning helps distinguish hand positions and object interactions. In IR, the benefits of reasoning are more dependent on detector capacity. The best IR performance is achieved when high-quality reasoning labels are combined with larger YOLO26 variants, suggesting that IR recognition requires both cleaner supervision and stronger feature capacity.

These results suggest that the main value of reasoning mode in DataDet is not to act as a direct online classifier, but to provide cleaner and more informative pseudo labels for downstream detector training. Its benefit is most evident when the standard annotation mode produces noisy labels and when the edge detector has sufficient capacity to exploit the improved supervision. Meanwhile, the Holo3-35B-A3B results show that higher label accuracy does not always translate linearly into better detector performance, indicating that label consistency, class-wise error distribution, and temporal stability of pseudo labels also play important roles in pseudo-label-based edge detector training.

C. Class-level Sensitivity to Pseudo-label Quality

Beyond the average performance reported in Table VIII, the per-class results in Tables IX–XVIII further show that different distracted-driving categories have different sensitivities to pseudo-label quality. Classes with clear visual patterns and stable spatial cues, such as `radio`, `phonecall_left`, and `phonecall_right`, remain relatively robust across annotation models. For example, under the RGB setting with YOLO26n, `radio` already achieves an F1-score of 92.1% using Gemma-4-26B-A4B in the standard mode and increases to 96.9% with Qwen3.6-35B-A3B in the reasoning mode. Similarly, `phonecall_right` improves from 91.3% to 97.7%, indicating that these classes can be learned effectively once the coarse action region is correctly identified.

In contrast, classes involving subtle hand-object interactions or visually ambiguous gestures are more sensitive to pseudo-label noise. The most evident case is `hair_and_makeup`. Under the RGB setting with YOLO26n, its F1-score is 82.7% with Gemma-4-26B-A4B in the standard mode, drops to 73.8% with Ovis2.6-30B-A3B in the standard mode, and rises to 91.5% with Qwen3.6-35B-A3B in the reasoning mode. A similar trend appears for `hold_on_the_wheel_right`, whose F1-score improves from 77.0% under Gemma-4-26B-A4B standard annotation to 92.5% under Qwen3.6-35B-A3B reasoning annotation. These results suggest that categories requiring fine-grained distinction between hands, steering wheel, face, and small objects benefit more from cleaner pseudo labels.

The class-level results also explain why average label accuracy alone cannot fully determine downstream detector performance. Although Qwen3.5-35B-A3B reasoning achieves the highest label accuracy on both RGB and IR pseudo labels, Qwen3.6-35B-A3B reasoning often produces the best RGB detector performance for YOLO26n and competitive results across larger YOLO26 variants. This indicates that pseudo-label quality should not only be measured by global correctness, but also by whether errors are concentrated in difficult classes. Errors in visually sensitive categories, such as `hair_and_makeup`, `texting_right`, and steering-wheel-related classes, have a stronger impact on downstream learning than errors in visually distinctive classes.

The IR results further highlight this class-level sensitivity. Compared with RGB, IR images provide weaker texture and color cues, making object-related behaviors harder to distinguish. For YOLO26n, Gemma-4-26B-A4B standard annotation yields only

22.7% F1-score for drinking and 34.4% for texting_left, while Qwen3.6-35B-A3B reasoning improves them to 75.6% and 76.3%, respectively. However, even with stronger annotation models, some IR classes remain noticeably lower than their RGB counterparts, especially hair_and_makeup and texting_right. This shows that pseudo-label quality is particularly important when the modality itself provides limited visual evidence.

These observations demonstrate that DataDet's gains are not uniformly distributed across classes. High-quality pseudo labels mainly improve the categories where visual ambiguity, hand-object overlap, and modality degradation make direct recognition difficult. Therefore, the effectiveness of cloud-side annotation should be evaluated not only by average label accuracy, but also by its ability to reduce class-specific noise in the most sensitive action categories.

D. RGB–IR Differences in Class-wise Detection

Tables IX–XVIII compare the class-wise behavior of DataDet under RGB and IR settings. The results show that the modality gap is highly class-dependent. RGB generally provides richer appearance cues, including color, texture, and object details, which are particularly useful for recognizing object-related distracted behaviors. In contrast, IR images preserve body structure and hand position but weaken fine-grained visual evidence, leading to larger performance drops for categories that depend on small objects or subtle appearance differences.

This difference is clear when comparing the strongest RGB and IR settings. With Qwen3.6-35B-A3B reasoning and YOLO26n, the average RGB F1-score reaches 94.6%, while the corresponding IR F1-score is 80.8%. The gap is especially large for texting_left and texting_right: their F1-scores decrease from 92.6% and 92.2% in RGB to 76.3% and 73.4% in IR, respectively. A similar degradation appears for drinking, which drops from 95.5% in RGB to 75.6% in IR. These classes require the detector to distinguish small objects such as phones or bottles and to capture detailed hand-object interactions, both of which become less reliable in IR images.

By contrast, classes dominated by spatial posture or hand-wheel geometry are more robust across modalities. For example, under Qwen3.6-35B-A3B reasoning with YOLO26n, radio achieves 96.9% F1-score in RGB and 98.2% in IR. The steering-wheel-related classes also remain relatively strong: hold_on_the_wheel_left obtains 95.9% in RGB and 84.4% in IR, while hold_on_the_wheel_right

obtains 92.5% in RGB and 80.1% in IR. Although there is still a visible decrease, these categories preserve more discriminative structure because they rely mainly on the relative position between hands and the steering wheel.

The RGB–IR gap is also reflected in false positive rates. Under weaker annotation settings, IR often produces much higher FPR for visually ambiguous classes. For example, with Gemma-4-26B-A4B standard annotation and YOLO26n, the IR FPR reaches 13.8% for `hair_and_makeup`, 9.9% for `hold_on_the_wheel_left`, and 11.3% for `hold_on_the_wheel_right`. In the RGB counterpart, the corresponding FPR values are much lower for most classes. This indicates that IR not only reduces recall for some object-related behaviors but can also increase confusion among visually similar hand and face actions when pseudo labels are noisy.

Larger edge models partially reduce the modality gap, but they do not remove it. For Qwen3.5-35B-A3B reasoning, increasing the edge detector from YOLO26n to YOLO26x improves IR F1-score from 81.5% to 87.6%, while RGB increases from 93.1% to 93.9%. This suggests that additional model capacity is more beneficial for IR, where the detector needs stronger feature extraction to compensate for weaker visual cues. However, even with YOLO26x, IR remains lower than RGB for several fine-grained categories, such as `texting_left`, `texting_right`, and `hair_and_makeup`.

These findings indicate that RGB and IR should not be treated as interchangeable modalities for distracted-driving detection. RGB is more suitable for recognizing object-centric behaviors, while IR is more reliable for structure-based actions but more vulnerable to fine-grained ambiguity. This also explains why high-quality cloud-side pseudo labels are especially important for IR training: when the visual signal is weaker, label noise in subtle categories is more likely to be amplified during downstream detector learning.

E. Detailed Results of Multi-round Detector Training

Tables XIX–XXIII and Tables XXIV–XXVIII provide detailed class-wise results across the multi-round training process. RGB contains 15 rounds from 600 to 9000 training samples, while IR contains 10 rounds from 600 to 6000 samples. In each update round, the newly added pseudo-labeled samples are kept approximately class-balanced. Figure 3 further shows the class prevalence of the accumulated RGB and IR training subsets, confirming that the closed-loop training process does not simply increase data

volume, but also maintains a controlled class distribution during data expansion.

For RGB, the most consistent gains appear in C1 and C2. With YOLO26n, C1 increases from 84.3% to 95.6% F1-score, and C2 increases from 74.8% to 92.6%. Similar improvements are also observed for larger models, showing that additional pseudo-labeled samples are effective for object-related and appearance-sensitive behaviors. C3, C5, C6, and C7 are already strong in early rounds. Their F1-scores are generally above 95% after several updates, which indicates that these classes have clearer visual patterns and are less dependent on later data expansion.

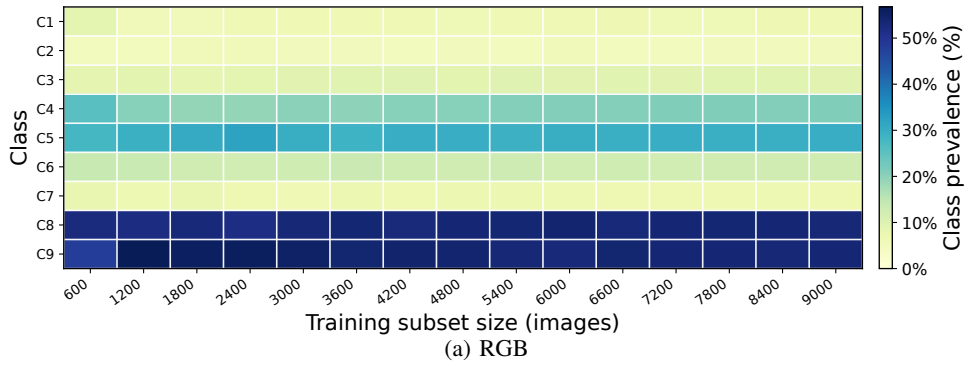
The difficult RGB cases are mainly C4, C8, and C9. C4 often has high recall but lower precision, suggesting that the detector tends to over-predict this class when right-hand wheel contact is visually ambiguous. For example, in YOLO26n, C4 reaches high recall across rounds, but its precision fluctuates from 89.0% at Round 600 to 72.7% at Round 1800 and 82.6% at Round 9000. C8 remains relatively stable, but its improvement is smaller than C1 and C2. C9 also fluctuates across rounds and model scales; in YOLO26x, it reaches 92.1% at Round 7800 but drops to 86.4% at Round 9000. These results suggest that right-side texting and hand-wheel states are sensitive to boundary cases and pseudo-label consistency.

For IR, class differences are more pronounced. C7 remains the easiest class, with F1-scores close to or above 98% for most models and rounds. C5 and C6 also maintain relatively strong performance, especially in later rounds, indicating that phone-call gestures and radio-related patterns can still be captured from coarse IR structure. In contrast, C1 and C2 are much harder in early rounds. With YOLO26n, C1 improves from 42.0% to 73.7%, and C2 improves from 41.4% to 62.0%. Although the gains are large, the final performance of C2 remains clearly lower than most other classes, showing that makeup-like behaviors are difficult to distinguish in IR due to weak texture and object cues.

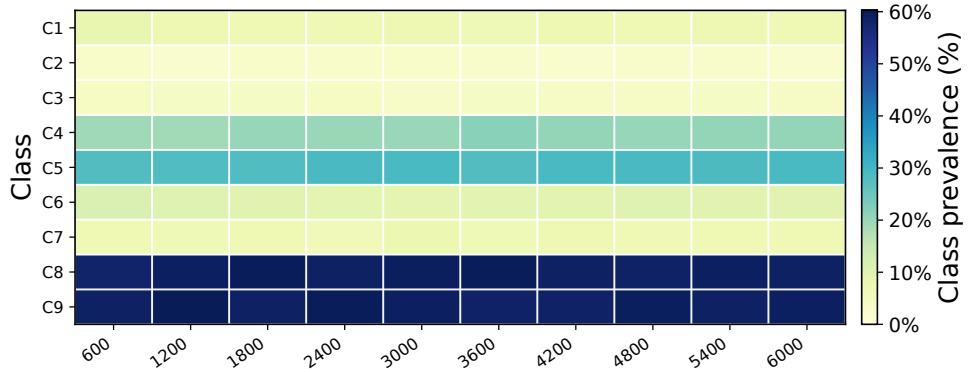
Several IR classes also show unstable behavior during later updates. C4 improves in some settings but remains sensitive to false positives. C8 and C9 benefit from additional data, but their gains are not strictly monotonic. For instance, YOLO26x improves C9 from 68.3% to 81.0%, but its performance varies across intermediate rounds. This indicates that additional pseudo-labeled data can improve coverage, but it may also introduce harder samples or residual label noise. The IR results therefore show both the strength and limitation of the closed-loop process: it can substantially improve weak classes, but fine-grained hand-object behaviors remain challenging when visual

evidence is limited.

These detailed results show that multi-round training does not improve all classes in the same way. Stable classes benefit mainly from maintaining high performance, while ambiguous classes benefit from additional samples but also show larger fluctuations. This suggests that later closed-loop updates should not only increase data scale, but also pay attention to class-wise pseudo-label quality and difficult boundary cases.



(a) RGB



(b) IR

Fig. 3: Class prevalence heatmaps of the constructed training subsets during closed-loop training on RGB and IR data. In each update round, newly added pseudo-labeled samples are approximately class-balanced. The heatmaps show the averaged class distribution of the accumulated training subsets across update rounds, where the x-axis denotes the training subset size and the y-axis denotes the distracted-driving classes.

References

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[2] A. Dontoh, S. Ivey, L. Sirbaugh, A. Danyo, and A. Aboah, "Visual dominance and emerging multimodal approaches in distracted driving detection: A review of machine learning techniques," *arXiv preprint arXiv:2505.01973*, 2025.

Method	Annotation Model	Inference Mode	Label Acc. (%)		Edge Model	Precision (%)		Recall (%)		F1-score (%)		FPR (%)	
			RGB	IR		RGB	IR	RGB	IR	RGB	IR	RGB	IR
Zero-shot	Gemma-4-26B-A4B	Standard	–	–	–	60.1	54.2	48.5	46.1	45.7	42.9	4.9	5.3
		Reasoning	–	–		65.9	60.0	55.5	47.7	57.5	51.1	4.1	4.3
	Holo3-35B-A3B	Standard	–	–		56.3	53.7	52.0	47.9	48.7	45.2	5.9	6.5
		Reasoning	–	–		60.5	54.2	57.8	47.8	56.2	45.3	5.1	6.3
	Ovis2.6-30B-A3B	Standard	–	–		57.3	53.6	57.7	51.8	56.4	48.1	5.0	5.5
		Reasoning	–	–		60.6	54.6	59.8	51.2	56.1	48.2	4.5	5.2
	Qwen3.5-35B-A3B	Standard	–	–		63.3	57.4	57.9	47.6	54.6	46.2	4.8	6.1
		Reasoning	–	–		71.6	69.3	59.4	55.3	61.0	60.0	3.4	3.5
	Qwen3.6-35B-A3B	Standard	–	–		63.7	57.2	56.2	48.7	57.4	49.8	4.5	5.3
		Reasoning	–	–		74.2	69.9	54.4	50.6	58.0	57.6	3.2	3.1
DataDet	Gemma-4-26B-A4B	Standard	70.2	67.3	YOLO26n	89.7	73.1	86.5	57.2	87.8	57.1	1.4	4.4
					YOLO26s	90.3	80.4	87.7	66.5	88.8	69.0	1.3	3.3
					YOLO26m	90.3	81.0	87.7	69.1	88.8	71.6	1.2	3.0
					YOLO26l	91.4	80.1	87.0	66.4	88.9	68.2	1.1	3.5
					YOLO26x	91.8	81.2	88.2	69.4	89.8	71.8	1.0	2.9
		Reasoning	90.2	81.1	YOLO26n	92.9	80.8	90.9	68.4	91.7	70.6	0.9	2.9
					YOLO26s	94.3	84.3	91.5	72.3	92.6	75.2	0.7	2.4
					YOLO26m	94.4	85.9	92.3	75.1	93.1	78.1	0.7	2.1
					YOLO26l	94.9	85.5	92.3	74.1	93.4	76.8	0.7	2.3
					YOLO26x	**96.2**	84.0	92.5	74.4	94.2	75.8	**0.5**	2.7
	Holo3-35B-A3B	Standard	85.3	77.2	YOLO26n	91.6	76.8	88.5	62.5	89.4	65.3	1.2	3.6
					YOLO26s	91.2	79.8	87.4	69.8	88.5	71.5	1.3	2.9
					YOLO26m	92.4	81.7	88.8	70.0	89.9	72.5	1.1	2.6
					YOLO26l	93.0	79.6	90.0	70.1	90.9	71.6	1.0	3.0
					YOLO26x	93.3	81.0	90.2	73.7	91.3	75.0	1.0	2.7
		Reasoning	89.5	84.6	YOLO26n	90.7	74.1	86.8	61.6	87.6	61.0	1.4	4.0
					YOLO26s	91.2	82.5	87.6	68.1	88.6	69.8	1.3	2.7
					YOLO26m	92.4	82.0	88.8	69.8	89.7	70.4	1.1	2.9
					YOLO26l	92.9	81.2	89.3	70.2	90.3	69.6	1.1	3.3
					YOLO26x	92.9	82.4	88.7	71.0	89.9	71.7	1.1	2.8
	Ovis2.6-30B-A3B	Standard	80.3	86.9	YOLO26n	87.1	82.8	79.4	74.7	81.0	76.5	2.4	2.2
					YOLO26s	88.8	88.5	82.2	79.7	84.0	82.4	1.9	1.5
					YOLO26m	89.1	88.4	83.5	80.6	85.0	82.7	1.8	1.5
					YOLO26l	88.6	89.5	82.9	81.4	84.4	83.7	1.9	1.4
					YOLO26x	89.8	88.5	84.2	81.1	85.8	83.0	1.6	1.5
		Reasoning	92.3	87.0	YOLO26n	92.5	82.8	90.0	74.7	90.5	76.5	1.1	2.2
					YOLO26s	94.0	88.5	91.0	79.7	92.1	82.4	0.8	1.5
					YOLO26m	93.6	88.4	91.0	80.6	91.9	82.7	0.9	1.5
					YOLO26l	94.2	89.5	91.7	81.4	92.6	83.7	0.8	1.4
					YOLO26x	94.1	90.6	91.5	79.3	92.4	83.0	0.8	1.2
	Qwen3.5-35B-A3B	Standard	83.5	83.1	YOLO26n	90.4	82.7	86.4	72.9	87.5	75.0	1.4	2.4
					YOLO26s	90.6	84.0	85.9	75.1	87.4	77.1	1.5	2.3
					YOLO26m	91.5	87.3	87.1	76.5	88.6	79.9	1.3	1.8
					YOLO26l	91.9	86.0	88.6	78.4	89.8	80.5	1.2	1.9
					YOLO26x	91.3	87.8	88.2	76.3	89.3	80.1	1.3	1.7
		Reasoning	**97.2**	**95.9**	YOLO26n	94.4	87.4	92.1	78.5	93.1	81.5	0.7	1.6
					YOLO26s	95.6	90.9	92.2	82.3	93.7	85.6	0.6	1.2
					YOLO26m	95.2	91.8	91.8	83.1	93.3	86.4	0.6	1.1
					YOLO26l	95.2	92.2	92.1	83.1	93.4	86.7	0.7	1.0
					YOLO26x	95.5	**92.8**	92.7	**84.2**	93.9	**87.6**	0.6	**0.9**
	Qwen3.6-35B-A3B	Standard	85.9	81.8	YOLO26n	89.9	80.9	84.9	71.9	86.1	73.5	1.6	2.7
					YOLO26s	92.3	84.1	88.3	73.4	89.5	76.2	1.1	2.2
					YOLO26m	92.8	84.1	88.9	72.4	90.2	75.8	1.0	2.3
					YOLO26l	92.8	80.6	89.1	75.3	90.2	76.6	1.1	2.7
					YOLO26x	93.2	83.1	88.7	74.6	90.2	76.9	1.0	2.3
		Reasoning	96.8	95.7	YOLO26n	95.7	87.3	**93.6**	77.8	**94.6**	80.8	**0.5**	1.6
					YOLO26s	95.3	90.1	92.4	81.4	93.7	84.5	0.6	1.3
					YOLO26m	95.4	91.3	91.9	81.4	93.3	85.2	0.7	1.1
					YOLO26l	95.5	91.2	91.8	82.3	93.4	85.7	0.6	1.1
					YOLO26x	**96.2**	90.8	92.6	82.4	94.1	85.7	**0.5**	1.1

TABLE VIII: Performance comparison among **DataDet**, zero-shot VLM baselines, and YOLO26 variant baselines on the DMD RGB and IR test sets. Label Acc. denotes the pseudo-label accuracy of each annotation model before training the edge detector. The second-best results are underlined, and the best results are highlighted in green bold.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	Gemma4-26B-A4B_Standard										**Gemma4-26B-A4B_Reasoning**										**Holo3-35B-A3B_Standard**									
Precision	81.7	86.8	88.7	70.5	96.7	89.2	98.0	98.1	97.4	89.7	96.0	79.4	92.5	85.6	97.7	96.9	94.2	97.8	96.3	92.9	96.9	98.7	86.3	71.0	95.7	91.5	89.6	98.0	97.1	91.6
Recall	94.4	79.0	81.0	84.8	88.4	93.5	86.9	84.0	86.6	86.5	94.1	85.9	97.8	95.5	77.4	98.3	93.3	85.9	90.2	90.9	91.4	73.0	97.2	96.4	93.5	97.7	89.7	76.6	81.2	88.5
F1-score	87.6	82.7	84.7	77.0	92.4	91.3	92.1	90.5	91.7	87.8	95.0	82.5	95.1	90.3	86.4	97.6	93.7	91.5	93.2	91.7	94.0	83.9	91.4	81.8	94.6	94.5	89.7	86.0	88.4	89.4
FPR	2.6	1.5	1.3	4.4	0.4	1.4	0.2	0.2	0.3	1.4	0.5	2.8	1.0	2.0	0.2	0.4	0.7	0.2	0.4	0.9	0.4	0.1	1.9	4.9	0.5	1.1	1.3	0.2	0.3	1.2
	Ovis2.6-30B-A3B_Standard										**Qwen3.5-35B-A3B_Standard**										**Qwen3.5-35B-A3B_Reasoning**									
Precision	91.0	95.7	84.4	44.9	95.9	78.4	99.6	97.4	96.5	87.1	86.9	97.4	90.2	62.4	95.6	87.5	99.5	97.6	96.5	90.4	97.3	95.2	91.9	82.7	99.4	96.4	99.9	98.5	88.5	94.4
Recall	83.6	60.0	89.9	89.6	92.0	91.1	62.1	80.8	65.3	79.4	94.0	68.2	92.4	91.4	91.2	90.0	92.2	87.4	71.0	86.4	86.9	85.5	98.6	98.1	95.0	98.2	92.3	87.5	87.0	92.1
F1-score	87.1	73.8	87.1	59.8	93.9	84.3	76.5	88.3	77.9	81.0	90.3	80.3	91.3	74.2	93.3	88.7	95.7	92.2	81.8	87.5	91.8	90.1	95.1	89.7	97.1	97.3	95.9	92.7	87.7	93.1
FPR	1.0	0.3	2.1	13.7	0.5	3.1	0.0	0.3	0.3	2.4	1.8	0.2	1.3	6.9	0.5	1.6	0.1	0.3	0.3	1.4	0.3	0.5	1.1	2.6	0.1	0.5	0.0	0.2	1.4	0.7
	Qwen3.6-35B-A3B_Standard										**Holo3-35B-A3B_Reasoning**										**Ovis2.6-30B-A3B__Reasoning**									
Precision	96.0	90.2	86.7	56.6	91.8	95.9	98.4	98.4	95.3	89.9	96.4	96.6	91.2	64.2	96.8	80.5	95.1	98.9	96.9	90.7	92.5	94.9	92.2	78.9	98.9	79.8	99.5	98.2	97.4	92.5
Recall	91.4	61.5	92.4	97.6	95.3	85.6	91.3	75.2	74.1	84.9	87.9	58.1	96.4	98.1	93.7	93.0	91.4	84.4	77.9	86.8	95.8	65.7	98.5	97.6	93.2	98.6	91.1	86.1	83.0	89.9
F1-score	93.6	73.1	89.5	71.7	93.5	90.4	94.7	85.3	83.3	86.1	92.0	72.6	93.7	77.6	95.2	86.3	93.2	91.1	86.4	87.6	94.1	77.7	95.2	87.2	96.0	88.2	95.1	91.8	89.6	90.5
FPR	0.5	0.8	1.8	9.3	1.1	0.5	0.2	0.2	0.5	1.6	0.4	0.3	1.2	6.8	0.4	2.8	0.6	0.1	0.3	1.4	1.0	0.4	1.0	3.3	0.1	3.1	0.1	0.2	0.3	1.1
	Qwen3.6-35B-A3B_Reasoning																													
Precision	97.7	93.7	94.0	87.4	99.1	97.2	99.9	97.9	94.5	95.7	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	93.5	89.4	97.9	98.3	93.2	98.2	94.2	87.8	90.0	93.6	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	95.5	91.5	95.9	92.5	96.1	97.7	96.9	92.6	92.2	94.6	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.3	0.8	0.8	1.8	0.1	0.3	0.0	0.2	0.7	0.5	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE IX: Per-class (C1–C9) and average performance of different annotation models using DataDet-YOLO26n on DMD under the **RGB** training and **RGB** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	Gemma4-26B-A4B_Standard										**Gemma4-26B-A4B_Reasoning**										**Holo3-35B-A3B_Standard**									
Precision	85.1	84.3	90.9	74.8	95.7	89.7	96.9	97.7	97.4	90.3	96.0	78.5	98.0	89.8	97.2	97.8	95.5	97.9	97.8	94.3	97.8	97.9	89.2	69.8	95.9	81.8	93.0	98.0	97.8	91.2
Recall	94.0	82.4	79.2	85.3	86.8	94.2	89.4	84.6	93.3	87.7	98.1	88.5	97.3	95.6	70.8	98.0	94.6	87.8	93.2	91.5	91.0	71.1	95.4	98.3	92.1	94.1	88.7	80.8	75.4	87.4
F1-score	89.3	83.3	84.7	79.7	91.0	91.9	93.0	90.7	95.3	88.8	97.1	83.2	97.7	92.6	81.9	97.9	95.0	92.5	95.5	92.6	94.3	82.4	92.2	81.6	93.9	87.5	90.8	88.6	85.2	88.5
FPR	2.1	1.9	1.0	3.6	0.5	1.3	0.4	0.2	0.3	1.3	0.5	3.0	0.2	1.3	0.3	0.3	0.6	0.2	0.3	0.7	0.3	0.2	1.5	5.3	0.5	2.6	0.8	0.2	0.2	1.3
	Ovis2.6-30B-A3B_Standard										**Qwen3.5-35B-A3B_Standard**										**Qwen3.5-35B-A3B_Reasoning**									
Precision	89.2	95.8	86.2	53.6	96.3	83.4	99.6	97.9	97.3	88.8	95.7	96.8	88.6	58.2	95.2	87.6	98.9	97.4	96.8	90.6	96.9	97.6	96.2	84.4	99.8	98.0	99.7	97.7	89.7	95.6
Recall	88.8	66.1	89.7	94.1	89.6	87.9	65.8	76.9	80.8	82.2	92.2	73.7	92.6	90.1	89.0	87.5	93.4	84.5	70.4	85.9	90.2	86.3	98.1	98.0	88.4	98.1	92.4	88.6	89.6	92.2
F1-score	89.0	78.2	88.0	68.3	92.9	85.6	79.3	86.1	88.3	84.0	93.9	83.7	90.6	70.7	92.0	87.6	96.1	90.5	81.5	87.4	93.4	91.6	97.1	90.7	93.7	98.1	95.9	92.9	89.6	93.7
FPR	1.3	0.4	1.8	10.2	0.4	2.2	0.0	0.2	0.3	1.9	0.5	0.3	1.5	8.1	0.6	1.5	0.1	0.3	0.3	1.5	0.4	0.3	0.5	2.3	0.0	0.2	0.0	0.3	1.3	0.6
	Qwen3.6-35B-A3B_Standard										**Holo3-35B-A3B_Reasoning**										**Ovis2.6-30B-A3B__Reasoning**									
Precision	93.7	94.9	87.5	69.0	94.9	98.2	98.1	97.3	96.7	92.3	97.1	93.8	88.5	69.1	96.5	83.6	95.8	98.9	97.6	91.2	94.9	95.2	95.6	84.6	99.3	83.1	99.0	98.0	96.6	94.0
Recall	95.1	63.8	95.3	97.4	89.6	94.6	91.6	77.4	89.7	88.3	89.6	66.3	97.0	98.1	93.0	92.1	92.6	78.1	81.4	87.6	96.2	68.2	98.2	97.1	90.1	97.0	95.5	86.6	89.8	91.0
F1-score	94.4	76.3	91.3	80.8	92.1	96.4	94.7	86.2	93.1	89.5	93.2	77.7	92.6	81.1	94.7	87.6	94.2	87.3	88.8	88.6	95.5	79.5	96.9	90.4	94.5	89.5	97.2	91.9	93.1	92.1
FPR	0.8	0.4	1.7	5.5	0.6	0.2	0.2	0.3	0.4	1.1	0.3	0.6	1.6	5.5	0.4	2.3	0.5	0.1	0.2	1.3	0.7	0.4	0.6	2.2	0.1	2.5	0.1	0.2	0.4	0.8
	Qwen3.6-35B-A3B_Reasoning																													
Precision	98.8	91.5	94.2	83.1	99.4	99.0	99.8	98.2	93.9	95.3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	92.4	91.6	97.6	98.3	93.2	97.7	91.8	85.6	83.6	92.4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	95.5	91.5	95.8	90.1	96.2	98.4	95.6	91.5	88.5	93.7	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.1	1.1	0.8	2.5	0.1	0.1	0.0	0.2	0.7	0.6	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE X: Per-class (C1–C9) and average performance of different annotation models using DataDet-YOLO26s on DMD under the **RGB** training and **RGB** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	Gemma4-26B-A4B_Standard										**Gemma4-26B-A4B_Reasoning**										**Holo3-35B-A3B_Standard**									
Precision	87.5	85.6	89.1	73.9	95.9	89.2	98.0	97.6	96.1	90.3	96.4	84.3	93.2	88.9	98.6	98.6	94.0	97.6	98.1	94.4	98.1	99.4	91.3	68.8	96.1	91.8	91.8	97.8	96.3	92.4
Recall	95.7	79.7	79.7	84.2	84.5	96.6	89.4	85.8	93.9	87.7	98.7	87.7	97.9	95.6	77.0	98.1	94.5	87.6	93.2	92.3	91.4	71.2	96.0	97.0	88.7	95.1	91.2	85.2	82.9	88.8
F1-score	91.4	82.5	84.1	78.8	89.9	92.7	93.5	91.3	95.0	88.8	97.6	85.9	95.5	92.1	86.5	98.4	94.3	92.3	95.6	93.1	94.6	83.0	93.6	80.5	92.3	93.5	91.5	91.1	89.1	89.9
FPR	1.7	1.7	1.2	3.7	0.4	1.5	0.2	0.3	0.5	1.2	0.5	2.0	0.9	1.5	0.1	0.2	0.8	0.3	0.2	0.7	0.2	0.1	1.1	5.5	0.4	1.1	1.0	0.2	0.4	1.1
	Ovis2.6-30B-A3B_Standard										**Qwen3.5-35B-A3B_Standard**										**Qwen3.5-35B-A3B_Reasoning**									
Precision	91.7	96.6	83.4	56.1	95.9	82.4	99.8	97.6	98.3	89.1	96.4	97.0	92.9	59.9	95.0	89.3	99.5	97.8	96.1	91.5	98.2	98.0	96.3	80.1	99.6	97.4	99.8	98.3	89.4	95.2
Recall	83.1	69.7	89.3	91.6	88.0	95.8	72.4	80.3	81.1	83.5	94.0	74.0	91.8	91.4	90.2	86.8	93.6	87.6	74.4	87.1	91.1	87.3	98.2	98.5	89.3	98.1	91.8	88.8	83.4	91.8
F1-score	87.2	81.0	86.3	69.6	91.8	88.6	83.9	88.1	88.9	85.0	95.2	84.0	92.3	72.4	92.5	88.0	96.5	92.4	83.9	88.6	94.5	92.3	97.2	88.3	94.2	97.8	95.6	93.3	86.3	93.3
FPR	0.9	0.3	2.2	9.0	0.5	2.6	0.0	0.2	0.2	1.8	0.4	0.3	0.9	7.6	0.6	1.3	0.1	0.2	0.4	1.3	0.2	0.2	0.5	3.1	0.0	0.3	0.0	0.2	1.2	0.6
	Qwen3.6-35B-A3B_Standard										**Holo3-35B-A3B_Reasoning**										**Ovis2.6-30B-A3B__Reasoning**									
Precision	96.5	95.0	89.6	70.6	93.5	96.4	99.1	97.3	97.0	92.8	97.7	97.7	88.7	67.9	97.0	89.1	97.3	97.8	98.7	92.4	97.4	93.7	94.5	82.1	99.1	82.7	97.9	98.2	96.9	93.6
Recall	95.7	69.1	92.7	97.5	86.6	97.3	93.2	78.4	89.3	88.9	92.0	65.4	97.0	97.4	90.9	97.0	93.2	83.0	82.9	88.8	96.2	70.4	97.7	96.9	91.2	98.6	91.7	87.3	88.6	91.0
F1-score	96.1	80.0	91.2	81.9	89.9	96.9	96.1	86.8	93.0	90.2	94.8	78.4	92.7	80.0	93.9	92.9	95.2	89.8	90.1	89.7	96.8	80.4	96.1	88.9	95.0	89.9	94.7	92.4	92.5	91.9
FPR	0.4	0.4	1.3	5.1	0.8	0.4	0.1	0.3	0.3	1.0	0.3	0.2	1.5	5.7	0.3	1.5	0.3	0.2	0.1	1.1	0.3	0.6	0.7	2.6	0.1	2.6	0.2	0.2	0.4	0.9
	Qwen3.6-35B-A3B_Reasoning																													
Precision	98.2	97.7	95.8	76.4	99.2	97.6	99.9	98.2	95.5	95.4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	96.5	88.3	98.2	97.9	90.8	98.5	93.1	87.9	76.1	91.9	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	97.3	92.8	97.0	85.8	94.8	98.0	96.4	92.8	84.7	93.3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.2	0.3	0.5	3.8	0.1	0.3	0.0	0.2	0.4	0.7	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XI: Per-class (C1–C9) and average performance of different annotation models using DataDet-YOLO26m on DMD under the **RGB** training and **RGB** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	Gemma4-26B-A4B_Standard										**Gemma4-26B-A4B_Reasoning**										**Holo3-35B-A3B_Standard**									
Precision	86.9	83.5	91.6	76.0	96.3	93.4	98.3	98.4	98.0	91.4	98.4	85.4	94.4	87.6	99.3	98.0	94.9	97.7	98.3	94.9	98.1	99.0	89.9	69.5	96.9	95.3	93.0	97.6	97.6	93.0
Recall	95.9	79.3	77.0	88.5	82.1	96.6	87.3	84.5	92.2	87.0	97.3	88.8	98.0	95.5	79.2	98.2	94.6	88.5	90.9	92.3	90.9	72.7	95.8	96.9	94.4	97.6	90.0	85.1	86.9	90.0
F1-score	91.2	81.3	83.7	81.7	88.6	95.0	92.5	90.9	95.0	88.9	97.8	87.1	96.2	91.4	88.1	98.1	94.8	92.9	94.5	93.4	94.3	83.9	92.7	81.0	95.6	96.4	91.5	91.0	91.9	90.9
FPR	1.8	2.0	0.9	3.5	0.4	0.9	0.2	0.2	0.2	1.1	0.2	1.9	0.7	1.7	0.1	0.2	0.6	0.3	0.2	0.7	0.2	0.1	1.3	5.3	0.4	0.6	0.8	0.3	0.3	1.0
	Ovis2.6-30B-A3B_Standard										**Qwen3.5-35B-A3B_Standard**										**Qwen3.5-35B-A3B_Reasoning**									
Precision	92.8	94.5	81.2	53.8	96.5	84.0	99.5	97.4	97.9	88.6	96.9	98.4	91.4	65.2	96.4	85.1	99.5	97.4	96.9	91.9	98.9	98.5	95.5	78.8	98.3	98.1	99.8	97.8	91.1	95.2
Recall	85.1	71.8	89.7	91.4	88.5	93.9	67.0	76.5	82.5	82.9	95.8	76.8	93.2	89.4	89.8	85.1	93.7	88.9	85.0	88.6	91.3	88.7	98.6	98.2	90.0	98.2	92.1	88.6	83.3	92.1
F1-score	88.8	81.6	85.3	67.7	92.3	88.7	80.1	85.7	89.6	84.4	96.3	86.2	92.3	75.4	93.0	85.1	96.5	93.0	90.6	89.8	94.9	93.4	97.0	87.5	94.0	98.2	95.8	93.0	87.1	93.4
FPR	0.8	0.5	2.6	9.8	0.4	2.2	0.0	0.3	0.2	1.9	0.4	0.2	1.1	6.0	0.4	1.9	0.1	0.3	0.3	1.2	0.1	0.2	0.6	3.3	0.2	0.2	0.0	0.2	1.0	0.7
	Qwen3.6-35B-A3B_Standard										**Holo3-35B-A3B_Reasoning**										**Ovis2.6-30B-A3B__Reasoning**									
Precision	97.1	93.9	86.9	67.9	96.8	97.3	99.6	97.7	97.6	92.8	98.1	98.8	90.7	68.9	97.1	89.8	97.1	97.7	97.8	92.9	97.4	96.1	92.4	83.7	99.2	85.0	99.4	97.9	97.2	94.2
Recall	96.3	65.4	92.3	97.8	89.9	95.8	93.8	80.2	89.9	89.1	93.4	67.2	97.4	96.8	91.5	98.0	92.6	81.8	84.9	89.3	97.5	72.8	98.4	96.4	90.5	98.6	93.5	84.6	93.3	91.7
F1-score	96.7	77.1	89.6	80.1	93.2	96.5	96.6	88.1	93.6	90.2	95.7	80.0	93.9	80.5	94.3	93.7	94.8	89.1	90.9	90.3	97.4	82.8	95.3	89.6	94.6	91.3	96.4	90.8	95.2	92.6
FPR	0.4	0.5	1.7	5.8	0.4	0.3	0.0	0.2	0.3	1.1	0.2	0.1	1.2	5.5	0.3	1.4	0.3	0.2	0.2	1.1	0.3	0.4	1.0	2.4	0.1	2.2	0.1	0.2	0.3	0.8
	Qwen3.6-35B-A3B_Reasoning																													
Precision	97.5	97.3	94.9	79.3	99.5	98.9	99.5	98.0	94.6	95.5	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	94.7	88.7	97.9	97.9	86.7	97.9	93.3	86.9	82.1	91.8	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	96.1	92.8	96.4	87.6	92.6	98.4	96.3	92.1	87.9	93.4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.3	0.3	0.7	3.2	0.1	0.1	0.1	0.2	0.6	0.6	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XII: Per-class (C1–C9) and average performance of different annotation models using DataDet-YOLO26l on DMD under the **RGB** training and **RGB** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	Gemma4-26B-A4B_Standard										**Gemma4-26B-A4B_Reasoning**										**Holo3-35B-A3B_Standard**									
Precision	87.8	88.0	95.1	79.6	95.9	89.9	95.9	97.4	97.1	91.8	97.5	89.9	97.8	90.0	99.3	98.6	96.1	98.3	98.2	96.2	97.8	99.3	92.1	71.0	96.8	92.4	95.3	97.8	97.5	93.3
Recall	98.0	81.7	78.8	86.2	84.2	95.9	91.5	84.0	93.8	88.2	98.3	87.2	97.7	96.0	79.3	98.2	93.6	88.7	93.7	92.5	94.0	76.0	95.9	97.2	92.4	96.3	91.7	86.2	82.3	90.2
F1-score	92.6	84.7	86.2	82.8	89.6	92.8	93.7	90.2	95.4	89.8	97.9	88.5	97.7	92.9	88.2	98.4	94.9	93.3	95.9	94.2	95.9	86.1	94.0	82.1	94.6	94.3	93.5	91.6	89.3	91.3
FPR	1.7	1.4	0.5	2.8	0.4	1.3	0.5	0.3	0.3	1.0	0.3	1.2	0.3	1.3	0.1	0.2	0.5	0.2	0.2	0.5	0.3	0.1	1.0	5.0	0.4	1.0	0.6	0.2	0.3	1.0
	Ovis2.6-30B-A3B_Standard										**Qwen3.5-35B-A3B_Standard**										**Qwen3.5-35B-A3B_Reasoning**									
Precision	93.3	96.6	87.1	58.1	96.8	82.6	99.7	96.9	97.6	89.8	97.5	98.7	88.3	63.8	95.0	86.1	98.3	97.7	96.2	91.3	98.4	97.6	94.0	80.6	99.5	99.3	99.8	97.7	93.0	95.5
Recall	83.3	69.7	90.8	92.0	89.5	96.3	70.6	79.1	86.2	84.2	93.5	76.9	92.3	88.5	89.6	85.1	93.8	88.5	85.5	88.2	93.3	89.6	98.5	98.2	90.9	98.1	93.3	87.3	85.4	92.7
F1-score	88.0	81.0	88.9	71.2	93.0	88.9	82.7	87.1	91.5	85.8	95.5	86.4	90.3	74.2	92.3	85.6	96.0	92.9	90.5	89.3	95.8	93.5	96.2	88.5	95.0	98.7	96.5	92.2	89.1	93.9
FPR	0.8	0.3	1.7	8.3	0.4	2.5	0.0	0.3	0.3	1.6	0.3	0.1	1.5	6.3	0.6	1.7	0.2	0.3	0.4	1.3	0.2	0.3	0.8	3.0	0.1	0.1	0.0	0.3	0.8	0.6
	Qwen3.6-35B-A3B_Standard										**Holo3-35B-A3B_Reasoning**										**Ovis2.6-30B-A3B__Reasoning**									
Precision	97.7	95.2	88.9	70.0	95.5	97.4	99.2	97.9	97.4	93.2	99.1	98.8	91.0	64.1	98.0	91.7	96.6	97.8	98.4	92.9	97.0	98.1	93.4	77.3	99.1	87.2	99.2	98.1	97.8	94.1
Recall	96.8	65.4	93.4	97.6	86.0	96.9	93.1	79.3	89.7	88.7	91.0	67.1	96.0	97.4	91.0	93.0	92.9	84.9	85.1	88.7	97.1	75.7	98.2	96.4	91.1	98.4	94.4	85.7	86.2	91.5
F1-score	97.3	77.5	91.1	81.5	90.5	97.2	96.1	87.6	93.4	90.2	94.9	80.0	93.5	77.3	94.4	92.4	94.7	90.9	91.3	89.9	97.1	85.5	95.7	85.8	94.9	92.4	96.8	91.5	91.6	92.4
FPR	0.3	0.4	1.5	5.2	0.5	0.3	0.1	0.2	0.3	1.0	0.1	0.1	1.2	6.8	0.2	1.0	0.4	0.2	0.2	1.1	0.4	0.2	0.9	3.5	0.1	1.8	0.1	0.2	0.2	0.8
	Qwen3.6-35B-A3B_Reasoning																													
Precision	98.1	97.7	97.2	79.0	99.5	98.7	99.9	98.1	97.3	96.2	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	97.0	89.9	98.1	98.0	87.9	98.3	94.1	88.6	81.1	92.6	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	97.6	93.7	97.7	87.5	93.4	98.5	96.9	93.1	88.5	94.1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.2	0.3	0.3	3.3	0.1	0.2	0.0	0.2	0.3	0.5	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XIII: Per-class (C1–C9) and average performance of different annotation models using DataDet-YOLO26x on DMD under the **RGB** training and **RGB** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	Gemma4-26B-A4B_Standard										**Gemma4-26B-A4B_Reasoning**										**Holo3-35B-A3B_Standard**									
Precision	73.8	35.0	52.6	47.4	98.0	70.1	94.9	95.1	91.0	73.1	93.7	42.7	79.1	64.5	88.0	81.3	81.4	98.9	97.3	80.8	95.6	67.6	70.0	40.1	81.7	74.8	68.5	98.3	94.2	76.8
Recall	13.4	59.5	88.3	81.5	66.7	54.6	94.1	21.0	35.6	57.2	43.0	61.2	90.4	85.4	76.9	73.4	96.1	57.3	32.0	68.4	59.2	44.6	79.1	84.6	71.7	69.2	72.1	51.4	30.7	62.5
F1-score	22.7	44.1	65.9	59.9	79.4	61.4	94.5	34.4	51.2	57.1	58.9	50.3	84.3	73.5	82.1	77.2	88.1	72.6	48.1	70.6	73.1	53.8	74.3	54.4	76.4	71.9	70.3	67.5	46.3	65.3
FPR	0.6	13.8	9.9	11.3	0.2	2.9	0.6	0.1	0.4	4.4	0.4	10.3	3.0	5.9	1.3	2.1	2.8	0.1	0.1	2.9	0.3	2.7	4.2	15.8	2.0	2.9	4.1	0.1	0.2	3.6
	Ovis2.6-30B-A3B_Standard										**Ovis2.6-30B-A3B_Reasoning**										**Qwen3.5-35B-A3B_Standard**									
Precision	89.9	66.7	78.0	81.9	93.1	60.3	83.6	98.3	93.8	82.8	89.9	66.7	78.0	81.9	93.1	60.3	83.6	98.3	93.8	82.8	95.1	76.0	76.8	58.8	89.0	72.8	82.7	97.9	95.6	82.7
Recall	56.1	35.1	90.6	86.1	84.6	92.7	97.8	58.4	70.9	74.7	56.1	35.1	90.6	86.1	84.6	92.7	97.8	58.4	70.9	74.7	53.7	55.1	84.9	90.1	84.3	85.6	95.3	58.1	48.6	72.9
F1-score	69.1	46.0	83.8	83.9	88.6	73.0	90.1	73.3	80.8	76.5	69.1	46.0	83.8	83.9	88.6	73.0	90.1	73.3	80.8	76.5	68.6	63.9	80.6	71.2	86.6	78.7	88.6	72.9	64.4	75.0
FPR	0.8	2.2	3.2	2.4	0.8	7.6	2.4	0.1	0.6	2.2	0.8	2.2	3.2	2.4	0.8	7.6	2.4	0.1	0.6	2.2	0.3	2.2	3.2	7.9	1.3	4.0	2.5	0.2	0.3	2.4
	Qwen3.5-35B-A3B_Reasoning										**Qwen3.6-35B-A3B_Standard**										**Holo3-35B-A3B_Reasoning**									
Precision	94.4	74.5	83.5	72.1	96.4	78.6	97.4	97.4	92.4	87.4	93.4	61.8	70.0	56.8	84.4	78.3	96.2	95.0	91.9	80.9	95.0	53.5	66.2	43.8	85.4	68.2	69.1	95.3	90.1	74.1
Recall	65.4	47.5	92.2	92.3	82.9	95.4	97.5	70.1	63.4	78.5	42.1	68.3	80.1	87.9	85.6	83.5	95.9	49.3	54.3	71.9	32.2	44.5	91.0	80.2	81.6	88.6	80.7	36.6	18.8	61.6
F1-score	77.3	58.0	87.6	81.0	89.2	86.2	97.4	81.5	75.2	81.5	58.0	64.9	74.7	69.0	85.0	80.8	96.0	64.9	68.3	73.5	48.1	48.5	76.6	56.7	83.5	77.1	74.5	52.9	31.1	61.0
FPR	0.5	2.0	2.3	4.5	0.4	3.2	0.3	0.2	0.7	1.6	0.4	5.3	4.3	8.4	2.0	2.9	0.5	0.3	0.6	2.7	0.2	4.8	5.8	12.8	1.7	5.2	4.5	0.2	0.3	4.0
	Qwen3.6-35B-A3B_Reasoning																													
Precision	96.0	76.9	78.3	70.9	98.2	79.1	99.7	97.6	89.3	87.3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	62.3	50.3	91.5	92.0	85.1	97.1	96.7	62.6	62.4	77.8	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	75.6	60.8	84.4	80.1	91.2	87.2	98.2	76.3	73.4	80.8	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.3	1.9	3.2	4.7	0.2	3.2	0.0	0.2	0.9	1.6	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XIV: Per-class (C1–C9) and average performance of different annotation models using DataDet-YOLO26n on DMD under the **IR** training and **IR** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	Gemma4-26B-A4B_Standard										**Gemma4-26B-A4B_Reasoning**										**Holo3-35B-A3B_Standard**									
Precision	86.8	43.2	62.6	57.6	93.5	88.5	98.4	97.2	96.2	80.4	91.5	49.0	84.4	69.8	90.9	91.6	84.1	99.2	98.5	84.3	93.3	76.3	74.6	51.3	89.2	72.1	73.8	97.2	90.4	79.8
Recall	43.7	75.7	81.6	85.3	71.4	55.2	95.2	40.2	49.8	66.5	50.4	78.2	90.7	88.2	64.0	76.3	96.9	57.4	48.2	72.3	56.2	43.2	84.5	85.0	82.4	84.2	93.5	57.2	41.8	69.8
F1-score	58.1	55.0	70.9	68.8	81.0	68.0	96.8	56.9	65.6	69.0	65.0	60.2	87.5	77.9	75.1	83.3	90.0	72.7	64.7	75.2	70.1	55.2	79.3	64.0	85.6	77.7	82.5	72.0	57.1	71.5
FPR	0.8	12.4	6.1	7.9	0.6	0.9	0.2	0.1	0.2	3.3	0.6	10.2	2.1	4.8	0.8	0.9	2.3	0.1	0.1	2.4	0.5	1.7	3.6	10.1	1.2	4.1	4.2	0.2	0.6	2.9
	Ovis2.6-30B-A3B_Standard										**Ovis2.6-30B-A3B_Reasoning**										**Qwen3.5-35B-A3B_Standard**									
Precision	85.3	88.8	86.6	80.1	98.4	74.6	90.3	98.3	94.0	88.5	85.3	88.8	86.6	80.1	98.4	74.6	90.3	98.3	94.0	88.5	95.2	88.5	81.4	58.4	84.0	73.4	86.1	95.2	93.8	84.0
Recall	82.7	45.2	90.8	90.8	82.1	97.1	96.6	71.3	60.7	79.7	82.7	45.2	90.8	90.8	82.1	97.1	96.6	71.3	60.7	79.7	63.7	57.6	84.0	88.6	85.2	91.6	95.2	62.1	47.4	75.1
F1-score	84.0	59.9	88.7	85.1	89.5	84.4	93.3	82.6	73.7	82.4	84.0	59.9	88.7	85.1	89.5	84.4	93.3	82.6	73.7	82.4	76.3	69.8	82.7	70.4	84.6	81.5	90.4	75.2	63.0	77.1
FPR	1.8	0.7	1.8	2.8	0.2	4.1	1.3	0.2	0.5	1.5	1.8	0.7	1.8	2.8	0.2	4.1	1.3	0.2	0.5	1.5	0.4	0.9	2.4	7.9	2.0	4.2	1.9	0.4	0.4	2.3
	Qwen3.5-35B-A3B_Reasoning										**Qwen3.6-35B-A3B_Standard**										**Holo3-35B-A3B_Reasoning**									
Precision	95.9	88.1	85.8	74.4	96.6	85.0	99.4	98.1	94.6	90.9	95.0	67.9	79.4	60.9	81.0	78.8	98.8	97.1	97.9	84.1	94.9	89.3	80.5	53.9	72.1	62.4	94.6	97.9	97.2	82.5
Recall	78.4	64.2	91.7	92.2	80.9	97.0	95.9	70.4	70.4	82.3	49.1	66.4	84.5	90.0	80.9	87.2	94.5	51.9	55.8	73.4	60.4	30.1	89.1	86.0	78.5	96.1	92.3	37.5	43.0	68.1
F1-score	86.3	74.3	88.7	82.3	88.1	90.6	97.6	82.0	80.7	85.6	64.7	67.1	81.9	72.6	81.0	82.8	96.6	67.7	71.1	76.2	73.8	45.0	84.6	66.3	75.2	75.7	93.4	54.3	59.6	69.8
FPR	0.4	1.1	1.9	4.0	0.4	2.1	0.1	0.2	0.5	1.2	0.3	3.9	2.7	7.2	2.4	2.9	0.1	0.2	0.1	2.2	0.4	0.4	2.7	9.2	3.8	7.2	0.7	0.1	0.2	2.7
	Qwen3.6-35B-A3B_Reasoning																													
Precision	93.2	93.0	86.9	74.3	99.3	80.2	99.9	97.9	86.2	90.1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	75.3	56.1	91.4	91.6	84.4	96.3	96.4	72.9	67.9	81.4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	83.3	70.0	89.1	82.0	91.2	87.5	98.1	83.6	75.9	84.5	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.7	0.5	1.7	4.0	0.1	3.0	0.0	0.2	1.4	1.3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XV: Per-class (C1–C9) and average performance of different annotation models using DataDet-YOLO26s on DMD under the **IR** training and **IR** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	Gemma4-26B-A4B_Standard										**Gemma4-26B-A4B_Reasoning**										**Holo3-35B-A3B_Standard**									
Precision	88.6	48.4	62.2	61.1	92.7	86.1	94.5	98.5	96.3	81.0	92.1	52.0	86.4	72.5	94.6	93.7	84.1	99.2	98.8	85.9	94.4	83.7	78.7	56.7	86.9	70.9	73.6	97.6	92.4	81.7
Recall	48.4	75.3	80.6	83.3	71.0	67.1	96.6	42.3	57.0	69.1	54.3	75.3	91.1	89.7	69.2	82.0	97.5	64.8	52.3	75.1	55.8	39.5	81.9	85.1	78.3	86.2	93.9	62.8	46.1	70.0
F1-score	62.6	58.9	70.2	70.5	80.4	75.5	95.6	59.2	71.6	71.6	68.3	61.5	88.7	80.2	79.9	87.5	90.3	78.4	68.4	78.1	70.2	53.7	80.2	68.1	82.4	77.8	82.5	76.5	61.5	72.5
FPR	0.8	10.0	6.1	6.6	0.7	1.3	0.7	0.1	0.3	3.0	0.6	8.7	1.8	4.3	0.5	0.7	2.3	0.1	0.1	2.1	0.4	1.0	2.8	8.1	1.5	4.4	4.2	0.2	0.5	2.6
	Ovis2.6-30B-A3B_Standard										**Ovis2.6-30B-A3B_Reasoning**										**Qwen3.5-35B-A3B_Standard**									
Precision	92.4	89.6	86.2	82.9	96.0	69.0	88.3	97.8	93.5	88.4	92.4	89.6	86.2	82.9	96.0	69.0	88.3	97.8	93.5	88.4	95.5	89.9	85.4	60.7	91.9	80.4	87.8	97.5	96.4	87.3
Recall	79.4	40.5	91.5	91.2	84.1	97.4	96.9	70.6	74.2	80.6	79.4	40.5	91.5	91.2	84.1	97.4	96.9	70.6	74.2	80.6	69.0	58.5	84.0	90.1	80.4	89.6	95.4	70.7	50.8	76.5
F1-score	85.4	55.8	88.8	86.8	89.6	80.8	92.4	82.0	82.7	82.7	85.4	55.8	88.8	86.8	89.6	80.8	92.4	82.0	82.7	82.7	80.1	70.9	84.7	72.5	85.7	84.8	91.5	82.0	66.5	79.9
FPR	0.8	0.6	1.8	2.4	0.4	5.5	1.6	0.2	0.6	1.5	0.8	0.6	1.8	2.4	0.4	5.5	1.6	0.2	0.6	1.5	0.4	0.8	1.8	7.3	0.9	2.7	1.7	0.2	0.2	1.8
	Qwen3.5-35B-A3B_Reasoning										**Qwen3.6-35B-A3B_Standard**										**Holo3-35B-A3B_Reasoning**									
Precision	93.3	95.5	88.6	74.0	94.6	89.4	99.0	98.3	93.7	91.8	97.4	74.0	79.5	55.6	80.2	78.9	97.0	97.3	96.6	84.1	96.5	87.6	79.6	55.3	70.1	64.0	89.8	96.7	98.3	82.0
Recall	85.3	64.6	92.0	92.3	80.6	97.3	96.0	71.8	67.8	83.1	51.3	64.6	81.5	89.6	77.8	76.9	95.3	52.4	62.2	72.4	61.3	40.8	91.4	88.0	83.3	96.7	95.9	34.2	37.0	69.8
F1-score	89.1	77.0	90.2	82.1	87.1	93.2	97.5	83.0	78.6	86.4	67.2	68.9	80.4	68.6	79.0	77.8	96.1	68.1	75.7	75.8	75.0	55.7	85.1	67.9	76.1	77.0	92.7	50.5	53.8	70.4
FPR	0.8	0.4	1.5	4.0	0.6	1.4	0.1	0.2	0.6	1.1	0.2	2.8	2.6	8.9	2.4	2.6	0.4	0.2	0.3	2.3	0.3	0.7	2.9	8.9	4.4	6.8	1.4	0.1	0.1	2.8
	Qwen3.6-35B-A3B_Reasoning																													
Precision	92.8	92.0	91.5	77.3	99.4	83.1	99.9	98.1	87.9	91.3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	78.1	53.2	90.8	91.9	75.8	96.8	96.0	77.0	72.5	81.4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	84.8	67.4	91.1	84.0	86.0	89.4	97.9	86.3	79.5	85.2	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.8	0.6	1.1	3.4	0.1	2.5	0.0	0.2	1.2	1.1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XVI: Per-class (C1–C9) and average performance of different annotation models using DataDet-YOLO26m on DMD under the **IR** training and **IR** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	Gemma4-26B-A4B_Standard										**Gemma4-26B-A4B_Reasoning**										**Holo3-35B-A3B_Standard**									
Precision	88.3	44.5	54.0	58.3	96.6	89.9	97.6	96.7	94.7	80.1	95.8	50.6	88.3	69.6	91.0	93.7	82.3	99.4	99.0	85.5	94.6	79.3	75.1	51.4	88.8	70.5	69.4	97.2	90.4	79.6
Recall	46.2	78.6	79.0	88.4	71.5	56.6	96.8	33.9	46.6	66.4	57.2	80.0	90.6	91.0	72.3	73.5	97.1	59.9	45.7	74.1	50.0	46.6	83.0	85.0	84.6	80.0	94.1	62.7	44.9	70.1
F1-score	60.7	56.8	64.2	70.3	82.2	69.5	97.2	50.2	62.5	68.2	71.6	62.0	89.5	78.9	80.6	82.4	89.1	74.7	62.6	76.8	65.4	58.7	78.8	64.0	86.6	75.0	79.9	76.3	60.0	71.6
FPR	0.8	12.2	8.4	7.9	0.3	0.8	0.3	0.1	0.3	3.5	0.3	9.7	1.5	5.0	0.9	0.6	2.6	0.0	0.1	2.3	0.4	1.5	3.4	10.1	1.3	4.2	5.2	0.2	0.6	3.0
	Ovis2.6-30B-A3B_Standard										**Ovis2.6-30B-A3B_Reasoning**										**Qwen3.5-35B-A3B_Standard**									
Precision	95.1	89.7	86.2	80.8	97.9	70.7	93.2	98.2	93.7	89.5	95.1	89.7	86.2	80.8	97.9	70.7	93.2	98.2	93.7	89.5	92.1	91.9	78.8	64.2	89.8	78.5	86.8	96.0	95.5	86.0
Recall	79.9	41.0	92.1	92.3	84.1	97.4	96.6	73.0	76.3	81.4	79.9	41.0	92.1	92.3	84.1	97.4	96.6	73.0	76.3	81.4	69.8	58.0	86.4	91.2	84.1	91.6	95.5	67.2	61.5	78.4
F1-score	86.9	56.3	89.0	86.1	90.5	81.9	94.9	83.7	84.1	83.7	86.9	56.3	89.0	86.1	90.5	81.9	94.9	83.7	84.1	83.7	79.4	71.1	82.4	75.4	86.8	84.6	90.9	79.1	74.8	80.5
FPR	0.5	0.6	1.8	2.7	0.2	5.1	0.9	0.2	0.6	1.4	0.5	0.6	1.8	2.7	0.2	5.1	0.9	0.2	0.6	1.4	0.8	0.6	2.9	6.3	1.2	3.1	1.8	0.3	0.4	1.9
	Qwen3.5-35B-A3B_Reasoning										**Qwen3.6-35B-A3B_Standard**										**Holo3-35B-A3B_Reasoning**									
Precision	95.2	95.8	89.5	74.7	96.4	88.5	99.5	97.8	92.6	92.2	90.5	69.8	76.4	56.3	81.4	74.3	97.6	84.5	94.7	80.6	93.5	92.2	75.5	46.8	74.3	66.0	89.5	96.7	96.7	81.2
Recall	84.9	63.8	91.7	92.4	79.9	96.8	95.8	73.0	69.7	83.1	52.5	74.9	77.6	90.3	86.1	66.0	96.2	63.3	70.8	75.3	60.7	42.6	91.4	92.0	88.4	94.4	93.6	42.2	26.5	70.2
F1-score	89.7	76.6	90.6	82.7	87.4	92.5	97.6	83.6	79.5	86.7	66.5	72.3	77.0	69.4	83.7	69.9	96.9	72.4	81.1	76.6	73.6	58.2	82.7	62.0	80.7	77.7	91.5	58.8	41.6	69.6
FPR	0.5	0.3	1.3	3.9	0.4	1.6	0.1	0.2	0.7	1.0	0.7	4.0	3.0	8.7	2.5	2.9	0.3	1.5	0.5	2.7	0.5	0.4	3.7	13.1	3.8	6.1	1.4	0.2	0.1	3.3
	Qwen3.6-35B-A3B_Reasoning																													
Precision	92.6	92.7	88.5	78.0	98.9	84.9	99.9	97.6	87.4	91.2	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	76.5	57.3	91.4	92.3	82.2	97.3	96.8	74.6	72.1	82.3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	83.8	70.9	90.0	84.6	89.8	90.7	98.3	84.6	79.0	85.7	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.8	0.6	1.5	3.3	0.1	2.2	0.0	0.2	1.3	1.1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XVII: Per-class (C1–C9) and average performance of different annotation models using DataDet-YOLO26l on DMD under the **IR** training and **IR** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	Gemma4-26B-A4B_Standard										**Gemma4-26B-A4B_Reasoning**										**Holo3-35B-A3B_Standard**									
Precision	87.5	48.7	62.6	60.4	94.8	85.2	96.4	98.0	97.0	81.2	94.4	44.8	81.8	74.9	94.3	88.6	78.7	99.4	98.9	84.0	92.3	87.7	73.4	55.5	88.7	73.0	76.9	96.4	84.8	81.0
Recall	52.4	75.1	80.8	83.7	74.2	67.4	97.6	40.1	52.8	69.4	47.2	82.1	91.0	91.1	76.0	76.8	98.4	60.1	47.3	74.4	63.7	50.3	83.2	83.6	85.9	90.4	94.7	62.7	49.2	73.7
F1-score	65.6	59.1	70.6	70.2	83.2	75.3	97.0	57.0	68.4	71.8	62.9	58.0	86.2	82.2	84.1	82.3	87.4	74.9	63.9	75.8	75.4	64.0	78.0	66.7	87.2	80.7	84.9	76.0	62.3	75.0
FPR	0.9	9.9	6.0	6.9	0.5	1.5	0.5	0.1	0.2	2.9	0.3	12.6	2.5	3.8	0.6	1.2	3.3	0.0	0.1	2.7	0.7	0.9	3.8	8.4	1.4	4.2	3.6	0.3	1.1	2.7
	Ovis2.6-30B-A3B_Standard										**Ovis2.6-30B-A3B_Reasoning**										**Qwen3.5-35B-A3B_Standard**									
Precision	93.1	90.6	85.6	81.8	96.1	68.9	90.2	98.2	92.0	88.5	95.0	92.2	85.8	84.3	97.2	74.0	91.7	98.7	96.1	90.6	95.6	92.2	85.0	63.9	91.7	79.9	89.6	97.2	94.8	87.8
Recall	81.9	39.1	91.4	91.8	84.1	96.0	97.4	72.5	75.4	81.1	81.9	41.7	90.5	91.1	72.9	97.3	96.8	67.8	73.9	79.3	73.4	54.6	84.2	89.1	75.3	91.7	96.0	68.1	54.1	76.3
F1-score	87.1	54.7	88.4	86.5	89.7	80.2	93.7	83.4	82.9	83.0	88.0	57.4	88.1	87.5	83.3	84.1	94.2	80.4	83.6	83.0	83.0	68.6	84.6	74.4	82.7	85.4	92.7	80.1	68.9	80.1
FPR	0.8	0.5	1.9	2.6	0.4	5.4	1.3	0.2	0.8	1.5	0.5	0.4	1.9	2.1	0.3	4.3	1.1	0.1	0.4	1.2	0.4	0.6	1.9	6.3	0.9	2.9	1.4	0.2	0.4	1.7
	Qwen3.5-35B-A3B_Reasoning										**Qwen3.6-35B-A3B_Standard**										**Holo3-35B-A3B_Reasoning**									
Precision	94.9	97.5	89.3	77.7	98.7	87.9	99.2	97.8	92.1	92.8	97.7	70.0	78.7	57.7	79.9	78.7	97.3	94.4	93.7	83.1	96.7	90.2	77.1	52.2	72.0	67.4	92.3	96.5	96.8	82.4
Recall	85.6	62.4	92.0	92.3	82.4	97.1	96.0	75.7	74.4	84.2	53.9	62.6	84.1	87.2	83.5	83.7	95.6	54.3	67.0	74.6	69.5	41.6	91.5	88.4	83.6	93.2	95.4	35.1	40.2	71.0
F1-score	90.0	76.1	90.6	84.4	89.8	92.3	97.6	85.3	82.3	87.6	69.4	66.1	81.3	69.5	81.7	81.1	96.4	68.9	78.1	76.9	80.9	56.9	83.7	65.6	77.4	78.2	93.9	51.5	56.8	71.7
FPR	0.6	0.2	1.4	3.3	0.1	1.7	0.1	0.2	0.8	0.9	0.2	3.3	2.8	8.0	2.6	2.8	0.3	0.4	0.6	2.3	0.3	0.6	3.4	10.1	4.1	5.6	1.0	0.2	0.2	2.8
	Qwen3.6-35B-A3B_Reasoning																													
Precision	93.4	88.2	86.8	78.5	99.0	85.5	99.6	97.6	88.8	90.8	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	76.1	59.3	91.6	92.3	81.8	97.3	96.3	73.2	73.7	82.4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	83.8	70.9	89.1	84.8	89.6	91.0	97.9	83.6	80.6	85.7	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.7	1.0	1.7	3.2	0.1	2.1	0.0	0.2	1.2	1.1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XVIII: Per-class (C1–C9) and average performance of different annotation models using DataDet-YOLO26x on DMD under the **IR** training and **IR** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	DataDet-YOLO26n, Round 600										**DataDet-YOLO26n, Round 1200**										**DataDet-YOLO26n, Round 1800**									
Precision	88.2	90.1	97.1	89.0	97.2	88.4	99.9	96.1	87.8	92.6	90.6	95.6	91.3	81.2	99.7	93.6	99.8	97.6	86.1	92.8	94.9	94.1	90.0	72.7	99.8	91.0	99.8	98.1	87.5	92.0
Recall	80.7	64.0	96.7	97.6	94.9	97.9	89.7	89.9	96.4	89.8	87.9	74.2	97.7	96.8	93.2	97.9	92.1	85.1	84.3	89.9	88.7	80.6	97.7	97.5	93.0	97.9	90.8	84.5	70.9	89.1
F1-score	84.3	74.8	96.9	93.1	96.0	92.9	94.5	92.9	91.9	90.8	89.3	83.5	94.4	88.3	96.4	95.7	95.8	91.0	85.2	91.1	91.7	86.9	93.7	83.3	96.3	94.4	95.1	90.8	78.3	90.0
FPR	1.3	0.9	0.4	1.5	0.3	1.6	0.0	0.5	1.7	0.9	1.1	0.4	1.2	2.8	0.0	0.8	0.0	0.3	1.7	0.9	0.6	0.6	1.4	4.6	0.0	1.2	0.0	0.2	1.3	1.1
	DataDet-YOLO26n, Round 2400										**DataDet-YOLO26n, Round 3000**										**DataDet-YOLO26n, Round 3600**									
Precision	96.9	93.6	91.6	83.7	99.4	97.3	99.9	97.5	86.1	94.0	96.0	95.3	92.4	81.3	99.6	97.2	99.7	98.3	91.8	94.6	97.1	95.4	91.3	80.5	99.9	98.4	99.8	97.5	89.6	94.4
Recall	84.7	86.3	97.7	97.6	94.1	97.8	93.4	85.8	87.0	91.6	91.9	84.3	98.2	97.7	94.5	98.1	92.5	87.8	84.3	92.2	90.6	86.9	97.7	97.9	93.1	98.0	92.2	85.5	84.0	91.8
F1-score	90.4	89.8	94.5	90.1	96.7	97.6	96.6	91.3	86.6	92.6	93.9	89.5	95.2	88.8	97.0	97.7	96.0	92.8	87.9	93.2	93.8	91.0	94.4	88.4	96.4	98.2	95.8	91.1	86.7	92.9
FPR	0.3	0.7	1.1	2.4	0.1	0.3	0.0	0.3	1.8	0.8	0.5	0.5	1.0	2.8	0.0	0.3	0.0	0.2	0.9	0.7	0.3	0.5	1.2	3.0	0.0	0.2	0.0	0.3	1.2	0.7
	DataDet-YOLO26n, Round 4200										**DataDet-YOLO26n, Round 4800**										**DataDet-YOLO26n, Round 5400**									
Precision	97.4	92.6	92.6	78.5	99.5	97.9	99.6	98.1	93.8	94.5	98.0	93.1	93.4	79.7	99.5	96.3	99.8	98.2	89.6	94.2	98.2	96.1	92.2	76.2	99.5	98.7	99.8	98.3	91.6	94.5
Recall	91.9	87.8	97.5	97.8	93.5	97.8	92.5	88.5	80.5	92.0	90.4	87.0	97.5	98.0	93.2	97.7	93.3	89.1	81.1	91.9	90.8	89.3	97.9	98.0	95.0	98.2	93.8	85.7	78.4	91.9
F1-score	94.6	90.1	95.0	87.1	96.4	97.9	95.9	93.0	86.6	93.0	94.1	90.0	95.4	87.9	96.2	97.0	96.5	93.4	85.1	92.8	94.3	92.6	95.0	85.7	97.2	98.5	96.7	91.6	84.5	92.9
FPR	0.3	0.9	1.0	3.3	0.1	0.3	0.0	0.2	0.7	0.7	0.2	0.8	0.9	3.1	0.1	0.5	0.0	0.2	1.2	0.8	0.2	0.4	1.0	3.8	0.1	0.2	0.0	0.2	0.9	0.8
	DataDet-YOLO26n, Round 6000										**DataDet-YOLO26n, Round 6600**										**DataDet-YOLO26n, Round 7200**									
Precision	97.0	96.5	92.3	82.9	99.0	98.0	99.9	97.6	92.2	95.0	96.5	95.9	94.8	86.0	99.6	95.2	99.9	98.1	94.2	95.6	97.0	96.5	91.3	86.1	99.5	96.1	99.4	98.2	92.2	95.1
Recall	93.1	88.6	98.2	97.7	93.6	98.0	93.4	87.1	85.5	92.8	91.4	88.0	98.1	98.0	94.3	98.4	93.0	89.5	89.4	93.4	92.4	86.9	98.2	97.7	91.6	98.2	93.7	86.6	90.0	92.8
F1-score	95.0	92.4	95.2	89.7	96.2	98.0	96.6	92.1	88.7	93.8	93.9	91.8	96.4	91.6	96.9	96.8	96.3	93.6	91.7	94.3	94.7	91.4	94.6	91.5	95.4	97.2	96.5	92.0	91.1	93.8
FPR	0.4	0.4	1.0	2.5	0.1	0.2	0.0	0.3	0.9	0.7	0.4	0.5	0.7	2.0	0.0	0.6	0.0	0.2	0.7	0.6	0.4	0.4	1.2	2.0	0.1	0.5	0.1	0.2	1.0	0.6
	DataDet-YOLO26n, Round 7800										**DataDet-YOLO26n, Round 8400**										**DataDet-YOLO26n, Round 9000**									
Precision	96.3	97.4	93.2	81.8	99.2	97.6	99.8	98.1	94.4	95.3	96.3	97.6	93.7	78.9	99.3	97.0	99.7	98.3	94.6	95.0	98.1	96.9	92.6	82.6	99.2	97.8	99.8	97.5	93.0	95.3
Recall	96.3	87.9	98.0	97.7	93.3	98.1	93.3	87.9	82.4	92.8	94.1	86.6	98.0	97.7	93.5	98.2	93.4	88.6	80.3	92.3	93.2	88.6	98.1	97.8	93.3	98.2	93.8	87.1	85.5	92.9
F1-score	96.3	92.4	95.6	89.1	96.2	97.8	96.5	92.7	88.0	93.8	95.2	91.8	95.8	87.3	96.3	97.6	96.5	93.2	86.9	93.4	95.6	92.6	95.3	89.6	96.2	98.0	96.7	92.0	89.1	93.9
FPR	0.5	0.3	0.9	2.7	0.1	0.3	0.0	0.2	0.6	0.6	0.4	0.3	0.8	3.3	0.1	0.4	0.0	0.2	0.6	0.7	0.2	0.3	1.0	2.6	0.1	0.3	0.0	0.3	0.8	0.6

TABLE XIX: Per-class (C1–C9) and average performance of DataDet-YOLO26n across 15 training rounds on DMD under the **RGB** training and **RGB** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	DataDet-YOLO26s, Round 600										**DataDet-YOLO26s, Round 1200**										**DataDet-YOLO26s, Round 1800**									
Precision	91.9	92.5	98.8	88.9	98.5	94.1	99.6	97.4	88.2	94.4	97.4	90.0	92.5	81.1	99.5	97.1	99.7	98.0	91.4	94.1	98.3	94.4	91.2	75.3	99.4	98.3	99.8	98.3	94.0	94.3
Recall	86.0	73.6	96.9	97.7	91.5	98.0	94.2	89.6	96.5	91.6	86.7	82.7	97.1	96.4	94.1	97.8	93.0	84.8	85.0	90.8	94.1	87.5	97.5	98.4	94.6	98.2	91.2	84.1	74.5	91.1
F1-score	88.9	82.0	97.9	93.1	94.9	96.0	96.8	93.3	92.2	92.8	91.7	86.2	94.8	88.1	96.8	97.4	96.2	90.9	88.1	92.2	96.1	90.8	94.2	85.3	97.0	98.2	95.3	90.6	83.1	92.3
FPR	0.9	0.7	0.1	1.5	0.2	0.8	0.0	0.3	1.6	0.7	0.3	1.1	1.0	2.8	0.1	0.4	0.0	0.2	1.0	0.8	0.2	0.7	1.2	4.0	0.1	0.2	0.0	0.2	0.6	0.8
	DataDet-YOLO26s, Round 2400										**DataDet-YOLO26s, Round 3000**										**DataDet-YOLO26s, Round 3600**									
Precision	96.8	90.4	92.7	77.6	99.4	99.5	99.9	97.8	92.5	94.1	97.8	96.1	95.6	77.5	99.4	98.6	99.8	98.3	95.7	95.4	97.4	94.6	95.2	75.5	99.6	99.5	99.9	98.1	91.5	94.6
Recall	88.2	88.7	97.2	98.3	93.2	97.9	93.8	83.7	77.9	91.0	94.1	87.3	97.9	97.9	91.3	98.3	91.5	86.9	81.0	91.8	91.1	87.6	97.3	98.0	91.4	97.9	92.8	86.1	75.7	90.9
F1-score	92.3	89.5	94.9	86.7	96.2	98.7	96.8	90.2	84.6	92.2	95.9	91.5	96.8	86.6	95.2	98.4	95.4	92.2	87.8	93.3	94.1	90.9	96.2	85.3	95.4	98.7	96.2	91.7	82.9	92.4
FPR	0.4	1.2	1.0	3.5	0.1	0.1	0.0	0.2	0.8	0.8	0.3	0.4	0.6	3.5	0.1	0.2	0.0	0.2	0.4	0.6	0.3	0.6	0.6	4.0	0.0	0.1	0.0	0.2	0.9	0.7
	DataDet-YOLO26s, Round 4200										**DataDet-YOLO26s, Round 4800**										**DataDet-YOLO26s, Round 5400**									
Precision	97.5	95.4	96.0	83.4	99.5	99.5	99.8	97.8	94.7	96.0	99.1	94.4	97.9	77.4	99.5	97.5	99.8	97.9	92.3	95.1	97.2	96.3	95.7	78.3	99.5	99.3	99.9	98.2	94.1	95.4
Recall	95.3	90.5	97.8	97.7	91.9	97.7	93.1	88.4	86.1	93.2	90.5	87.2	97.9	98.3	93.1	98.3	93.0	89.6	78.7	91.8	94.9	87.8	97.7	98.0	90.9	98.1	93.3	88.3	80.6	92.2
F1-score	96.4	92.9	96.9	90.0	95.6	98.6	96.3	92.9	90.2	94.4	94.6	90.6	97.9	86.6	96.2	97.9	96.3	93.6	84.9	93.2	96.0	91.9	96.7	87.1	95.0	98.7	96.5	93.0	86.8	93.5
FPR	0.3	0.5	0.5	2.4	0.1	0.1	0.0	0.2	0.6	0.5	0.1	0.7	0.3	3.6	0.1	0.3	0.0	0.2	0.8	0.7	0.3	0.4	0.6	3.4	0.1	0.1	0.0	0.2	0.6	0.6
	DataDet-YOLO26s, Round 6000										**DataDet-YOLO26s, Round 6600**										**DataDet-YOLO26s, Round 7200**									
Precision	98.1	94.3	93.3	81.9	99.3	99.3	99.8	97.5	91.8	95.0	98.5	93.3	94.7	82.7	99.4	99.4	99.7	98.4	90.5	95.2	96.1	96.8	93.4	76.8	99.5	97.4	99.8	98.3	96.5	95.0
Recall	92.3	90.3	97.6	98.1	92.6	97.9	92.0	86.0	83.9	92.3	89.2	90.1	97.9	97.5	91.7	98.0	92.9	86.0	85.0	92.0	95.3	86.8	97.7	98.1	90.7	98.2	92.9	84.4	77.2	91.3
F1-score	95.1	92.2	95.4	89.3	95.9	98.6	95.7	91.4	87.7	93.5	93.6	91.7	96.3	89.5	95.4	98.7	96.2	91.7	87.7	93.4	95.7	91.5	95.5	86.2	94.9	97.8	96.2	90.8	85.8	92.7
FPR	0.2	0.7	0.9	2.7	0.1	0.1	0.0	0.3	0.9	0.7	0.2	0.8	0.7	2.5	0.1	0.1	0.0	0.2	1.1	0.6	0.5	0.4	0.9	3.7	0.1	0.3	0.0	0.2	0.3	0.7
	DataDet-YOLO26s, Round 7800										**DataDet-YOLO26s, Round 8400**										**DataDet-YOLO26s, Round 9000**									
Precision	98.1	98.7	93.5	81.4	99.6	97.7	99.8	97.7	95.5	95.8	98.4	96.9	95.2	82.4	99.5	99.6	99.6	98.3	91.9	95.8	98.4	98.4	95.7	83.6	99.2	98.6	99.9	97.5	93.6	96.1
Recall	94.7	87.4	97.8	97.9	90.5	98.7	93.1	85.6	85.8	92.4	93.6	89.2	97.7	97.8	92.1	97.7	93.6	86.6	85.1	92.6	95.4	87.5	98.0	97.8	90.5	98.2	92.9	86.5	88.0	92.8
F1-score	96.4	92.7	95.6	88.9	94.9	98.2	96.3	91.3	90.4	93.9	95.9	92.9	96.4	89.5	95.7	98.7	96.5	92.1	88.4	94.0	96.9	92.6	96.8	90.2	94.6	98.4	96.3	91.7	90.7	94.2
FPR	0.2	0.1	0.8	2.8	0.0	0.3	0.0	0.2	0.5	0.6	0.2	0.4	0.6	2.6	0.1	0.0	0.0	0.2	0.9	0.6	0.2	0.2	0.6	2.4	0.1	0.2	0.0	0.3	0.8	0.5

TABLE XX: Per-class (C1–C9) and average performance of DataDet-YOLO26s across 15 training rounds on DMD under the **RGB** training and **RGB** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	DataDet-YOLO26m, Round 600										**DataDet-YOLO26m, Round 1200**										**DataDet-YOLO26m, Round 1800**									
Precision	92.3	90.9	98.8	85.7	99.2	94.8	99.8	97.4	91.8	94.5	98.3	89.9	97.0	91.5	98.5	97.7	99.6	98.0	90.8	95.7	98.4	93.5	95.7	83.0	99.0	97.4	99.9	98.0	94.6	95.5
Recall	90.0	71.4	96.9	97.1	88.2	97.9	93.3	89.6	91.1	90.6	87.3	84.0	96.9	97.7	84.0	97.8	94.2	86.3	95.3	91.5	90.2	86.8	97.0	97.8	87.3	98.3	91.5	86.0	88.3	91.5
F1-score	91.2	80.0	97.9	91.1	93.4	96.3	96.4	93.3	91.5	92.3	92.5	86.8	97.0	94.5	90.7	97.8	96.8	91.8	93.0	93.4	94.1	90.0	96.4	89.8	92.8	97.8	95.5	91.6	91.3	93.3
FPR	0.9	0.9	0.1	2.0	0.1	0.7	0.0	0.3	1.0	0.7	0.2	1.2	0.4	1.1	0.2	0.3	0.0	0.2	1.2	0.5	0.2	0.8	0.5	2.5	0.1	0.3	0.0	0.2	0.6	0.6
	DataDet-YOLO26m, Round 2400										**DataDet-YOLO26m, Round 3000**										**DataDet-YOLO26m, Round 3600**									
Precision	98.3	94.2	94.2	75.9	99.4	98.5	99.9	97.9	89.5	94.2	98.1	94.5	97.6	79.9	99.6	96.8	99.7	98.2	95.6	95.5	97.3	96.2	97.6	85.0	99.5	97.6	99.9	97.9	90.5	95.7
Recall	87.7	87.8	97.5	97.7	86.4	98.1	93.3	85.5	76.2	90.0	91.4	86.0	98.0	97.6	83.3	98.6	93.6	88.8	83.3	91.2	92.3	86.0	97.2	97.7	86.7	98.0	92.3	87.5	88.5	91.8
F1-score	92.7	90.9	95.8	85.4	92.4	98.3	96.5	91.3	82.4	91.7	94.6	90.0	97.8	87.8	90.7	97.7	96.6	93.3	89.0	93.1	94.8	90.8	97.4	91.0	92.6	97.8	95.9	92.4	89.5	93.6
FPR	0.2	0.7	0.8	3.9	0.1	0.2	0.0	0.2	1.1	0.8	0.2	0.6	0.3	3.1	0.0	0.4	0.0	0.2	0.5	0.6	0.3	0.4	0.3	2.1	0.1	0.3	0.0	0.2	1.2	0.5
	DataDet-YOLO26m, Round 4200										**DataDet-YOLO26m, Round 4800**										**DataDet-YOLO26m, Round 5400**									
Precision	99.1	92.5	98.6	85.1	99.6	99.5	99.9	97.6	94.3	96.3	97.7	96.3	97.8	85.1	99.4	97.9	99.7	98.1	94.8	96.3	98.7	95.0	96.1	82.2	98.9	99.2	99.9	97.6	92.9	95.6
Recall	92.3	90.5	97.3	97.8	86.3	97.8	92.3	89.7	88.6	92.5	94.9	86.9	97.1	98.0	86.2	97.8	93.9	88.3	89.6	92.5	92.3	88.0	97.5	97.6	88.1	98.1	92.9	88.2	86.0	92.1
F1-score	95.6	91.5	98.0	91.0	92.5	98.6	95.9	93.5	91.4	94.2	96.3	91.4	97.5	91.1	92.3	97.9	96.7	92.9	92.1	94.2	95.4	91.4	96.8	89.3	93.2	98.6	96.3	92.7	89.3	93.7
FPR	0.1	0.9	0.2	2.1	0.0	0.1	0.0	0.3	0.7	0.5	0.3	0.4	0.3	2.1	0.1	0.3	0.0	0.2	0.6	0.5	0.1	0.6	0.5	2.6	0.1	0.1	0.0	0.3	0.8	0.6
	DataDet-YOLO26m, Round 6000										**DataDet-YOLO26m, Round 6600**										**DataDet-YOLO26m, Round 7200**									
Precision	99.3	93.6	97.1	81.0	99.3	98.7	99.9	97.6	95.3	95.8	97.5	95.3	97.2	77.7	99.3	99.1	99.9	97.8	92.9	95.2	98.3	96.3	95.6	77.6	99.1	98.4	99.9	98.3	94.5	95.3
Recall	93.2	90.1	97.7	97.6	86.4	97.1	92.7	88.1	84.1	91.9	93.1	88.1	97.7	98.0	86.0	96.9	92.7	88.6	78.6	91.1	92.5	87.5	97.7	97.8	83.5	98.3	92.9	86.8	80.7	90.9
F1-score	96.1	91.8	97.4	88.5	92.4	97.9	96.2	92.6	89.3	93.6	95.2	91.6	97.4	86.7	92.1	98.0	96.2	93.0	85.1	92.8	95.3	91.7	96.6	86.5	90.7	98.3	96.3	92.2	87.1	92.7
FPR	0.1	0.8	0.4	2.9	0.1	0.2	0.0	0.3	0.5	0.6	0.3	0.5	0.3	3.5	0.1	0.1	0.0	0.2	0.8	0.7	0.2	0.4	0.6	3.5	0.1	0.2	0.0	0.2	0.6	0.6
	DataDet-YOLO26m, Round 7800										**DataDet-YOLO26m, Round 8400**										**DataDet-YOLO26m, Round 9000**									
Precision	98.3	97.5	93.6	75.9	99.4	97.6	99.9	97.9	94.6	95.0	98.3	95.2	96.8	79.5	99.3	99.0	99.9	97.8	94.2	95.6	98.2	97.1	95.2	82.3	99.4	98.4	99.9	98.1	96.0	96.1
Recall	95.5	87.7	97.9	97.8	89.3	98.3	92.8	87.7	75.9	91.4	93.8	90.1	97.9	97.9	88.1	98.5	93.3	88.3	80.4	92.0	97.3	87.4	97.7	97.9	87.2	97.6	93.3	86.1	86.1	92.3
F1-score	96.9	92.3	95.7	85.5	94.1	97.9	96.2	92.5	84.2	92.8	96.0	92.6	97.4	87.7	93.4	98.7	96.5	92.8	86.7	93.5	97.7	92.0	96.4	89.4	92.9	98.0	96.5	91.7	90.8	93.9
FPR	0.2	0.3	0.8	3.9	0.1	0.3	0.0	0.2	0.5	0.7	0.2	0.6	0.4	3.2	0.1	0.1	0.0	0.2	0.6	0.6	0.2	0.3	0.6	2.6	0.1	0.2	0.0	0.2	0.4	0.5

TABLE XXI: Per-class (C1–C9) and average performance of DataDet-YOLO26m across 15 training rounds on DMD under the **RGB** training and **RGB** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	DataDet-YOLO26l, Round 600										**DataDet-YOLO26l, Round 1200**										**DataDet-YOLO26l, Round 1800**									
Precision	91.5	88.2	99.6	84.0	99.3	96.6	99.9	97.3	88.8	93.9	97.9	95.1	95.2	83.9	99.6	96.3	99.6	97.1	91.9	95.2	97.5	94.2	94.4	76.1	99.4	97.8	99.8	97.6	94.6	94.6
Recall	83.8	73.4	96.8	97.6	87.1	97.8	92.4	89.4	87.4	89.5	90.8	80.1	97.3	97.4	85.5	97.7	92.8	85.4	88.7	90.6	92.3	86.4	97.4	97.8	90.2	98.3	91.6	85.1	76.2	90.6
F1-score	87.5	80.1	98.2	90.3	92.8	97.2	96.0	93.2	88.1	91.5	94.2	87.0	96.3	90.1	92.0	97.0	96.1	90.9	90.3	92.7	94.9	90.1	95.9	85.6	94.6	98.0	95.5	90.9	84.4	92.2
FPR	1.0	1.2	0.0	2.3	0.1	0.4	0.0	0.3	1.4	0.8	0.2	0.5	0.6	2.3	0.0	0.5	0.0	0.3	1.0	0.6	0.3	0.7	0.7	3.8	0.1	0.3	0.0	0.3	0.5	0.7
	DataDet-YOLO26l, Round 2400										**DataDet-YOLO26l, Round 3000**										**DataDet-YOLO26l, Round 3600**									
Precision	99.1	90.5	93.2	87.8	99.7	99.4	99.9	98.2	90.1	95.3	98.2	95.8	95.5	77.8	99.5	98.2	99.7	98.0	91.4	94.9	98.8	93.8	96.1	78.0	99.4	99.4	99.9	98.0	90.6	94.9
Recall	83.7	89.9	97.5	98.0	87.1	97.8	95.2	83.9	92.2	91.7	88.9	86.7	97.7	98.3	89.0	98.2	92.8	86.3	80.8	91.0	89.4	87.8	97.7	97.9	87.5	97.8	93.3	87.3	79.0	90.9
F1-score	90.8	90.2	95.3	92.6	93.0	98.6	97.5	90.5	91.1	93.3	93.3	91.0	96.6	86.8	94.0	98.2	96.1	91.8	85.8	92.6	93.9	90.7	96.9	86.9	93.1	98.6	96.5	92.3	84.4	92.6
FPR	0.1	1.2	0.9	1.7	0.0	0.1	0.0	0.2	1.3	0.6	0.2	0.5	0.6	3.5	0.1	0.2	0.0	0.2	1.0	0.7	0.1	0.7	0.5	3.4	0.1	0.1	0.0	0.2	1.0	0.7
	DataDet-YOLO26l, Round 4200										**DataDet-YOLO26l, Round 4800**										**DataDet-YOLO26l, Round 5400**									
Precision	97.7	92.8	98.5	80.3	99.6	99.0	99.9	98.1	97.0	95.9	97.2	96.3	98.3	78.6	99.5	99.1	99.8	98.0	94.4	95.7	98.4	94.9	95.7	83.7	99.6	98.1	99.9	97.8	94.6	95.9
Recall	93.1	88.5	97.5	98.2	88.9	97.8	92.8	89.3	82.3	92.0	94.4	86.7	97.2	98.0	87.9	97.9	93.4	89.0	80.2	91.6	92.1	89.1	97.1	97.9	87.7	98.7	93.5	88.5	87.7	92.5
F1-score	95.3	90.6	98.0	88.4	94.0	98.4	96.2	93.5	89.0	93.7	95.8	91.2	97.7	87.3	93.4	98.5	96.5	93.3	86.7	93.4	95.1	91.9	96.4	90.3	93.3	98.4	96.6	92.9	91.0	94.0
FPR	0.3	0.9	0.2	3.0	0.0	0.1	0.0	0.2	0.3	0.6	0.3	0.4	0.2	3.3	0.1	0.1	0.0	0.2	0.6	0.6	0.2	0.6	0.5	2.4	0.0	0.2	0.0	0.2	0.6	0.5
	DataDet-YOLO26l, Round 6000										**DataDet-YOLO26l, Round 6600**										**DataDet-YOLO26l, Round 7200**									
Precision	98.8	96.6	95.2	80.8	99.4	99.2	99.9	97.9	93.9	95.7	98.3	95.4	96.5	78.6	99.3	99.5	99.8	97.8	93.7	95.4	98.7	95.0	92.7	77.6	99.5	97.9	99.9	98.1	95.6	95.0
Recall	93.4	89.0	97.9	98.2	89.6	98.2	91.9	87.0	84.0	92.2	92.4	89.6	97.9	98.0	87.1	97.8	93.5	88.3	80.2	91.7	94.3	89.2	98.0	97.9	90.6	98.2	93.4	86.1	78.7	91.8
F1-score	96.0	92.6	96.5	88.6	94.3	98.7	95.7	92.1	88.6	93.7	95.3	92.4	97.2	87.3	92.8	98.6	96.6	92.8	86.4	93.3	96.5	92.0	95.3	86.6	94.9	98.1	96.6	91.7	86.3	93.1
FPR	0.1	0.4	0.6	2.9	0.1	0.1	0.0	0.2	0.7	0.6	0.2	0.5	0.4	3.3	0.1	0.1	0.0	0.2	0.7	0.6	0.2	0.6	1.0	3.5	0.1	0.3	0.0	0.2	0.4	0.7
	DataDet-YOLO26l, Round 7800										**DataDet-YOLO26l, Round 8400**										**DataDet-YOLO26l, Round 9000**									
Precision	98.0	98.8	90.7	81.9	99.5	99.2	99.9	97.9	93.2	95.5	98.2	96.6	97.2	79.5	99.2	98.3	99.9	98.6	95.3	95.9	98.1	90.9	90.2	86.4	99.5	99.4	99.9	98.1	95.4	95.3
Recall	95.3	89.4	98.2	97.7	90.5	98.2	94.7	85.1	85.3	92.7	95.5	89.9	98.0	97.9	90.7	98.6	93.8	88.6	80.1	92.6	91.7	92.3	98.1	98.5	93.2	98.1	93.2	86.4	88.1	93.3
F1-score	96.6	93.9	94.3	89.1	94.8	98.7	97.2	91.0	89.1	93.9	96.9	93.1	97.6	87.8	94.8	98.4	96.8	93.3	87.0	94.0	94.8	91.6	94.0	92.0	96.3	98.7	96.4	91.9	91.6	94.2
FPR	0.2	0.1	1.3	2.7	0.1	0.1	0.0	0.2	0.8	0.6	0.2	0.4	0.3	3.1	0.1	0.2	0.0	0.2	0.5	0.6	0.2	1.1	1.3	1.9	0.1	0.1	0.0	0.2	0.5	0.6

TABLE XXII: Per-class (C1–C9) and average performance of DataDet-YOLO26l across 15 training rounds on DMD under the **RGB** training and **RGB** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	DataDet-YOLO26x, Round 600										**DataDet-YOLO26x, Round 1200**										**DataDet-YOLO26x, Round 1800**									
Precision	94.1	93.1	99.6	90.0	99.2	92.8	99.9	97.3	88.8	95.0	96.4	95.6	96.0	80.4	99.8	97.4	99.3	97.3	89.6	94.7	96.6	96.6	94.4	73.1	99.7	98.9	99.9	98.1	95.5	94.8
Recall	86.0	71.4	97.1	97.7	85.7	98.1	94.9	89.1	95.0	90.5	90.3	80.8	97.2	96.9	83.1	97.9	92.3	86.0	82.0	89.6	95.6	86.1	97.6	98.0	85.9	98.4	92.3	85.9	70.7	90.0
F1-score	89.8	80.8	98.4	93.7	91.9	95.4	97.3	93.0	91.8	92.5	93.3	87.6	96.6	87.9	90.7	97.7	95.7	91.3	85.6	91.8	96.1	91.1	96.0	83.7	92.3	98.6	95.9	91.6	81.3	91.8
FPR	0.7	0.7	0.0	1.4	0.1	0.9	0.0	0.3	1.5	0.6	0.4	0.5	0.5	3.0	0.0	0.3	0.1	0.3	1.2	0.7	0.4	0.4	0.7	4.5	0.0	0.1	0.0	0.2	0.4	0.8
	DataDet-YOLO26x, Round 2400										**DataDet-YOLO26x, Round 3000**										**DataDet-YOLO26x, Round 3600**									
Precision	97.9	93.6	94.2	78.5	99.7	99.5	99.8	97.4	90.7	94.6	97.8	94.8	98.0	78.5	99.4	98.8	99.8	97.9	94.9	95.6	98.1	96.1	94.8	77.1	99.6	99.5	100.0	98.1	91.9	95.0
Recall	88.5	86.6	97.3	98.0	86.4	97.9	93.4	85.3	80.3	90.4	92.5	85.8	97.8	97.3	86.9	98.3	92.5	88.8	80.4	91.2	91.2	87.6	97.3	98.0	83.9	98.0	92.3	86.7	78.9	90.4
F1-score	93.0	89.9	95.7	87.2	92.6	98.7	96.5	91.0	85.2	92.2	95.1	90.1	97.9	86.9	92.8	98.6	96.0	93.2	87.0	93.1	94.5	91.7	96.0	86.3	91.1	98.8	96.0	92.0	84.9	92.4
FPR	0.2	0.7	0.7	3.4	0.0	0.1	0.0	0.3	1.0	0.7	0.3	0.6	0.2	3.3	0.1	0.1	0.0	0.2	0.5	0.6	0.2	0.4	0.7	3.6	0.0	0.1	0.0	0.2	0.9	0.7
	DataDet-YOLO26x, Round 4200										**DataDet-YOLO26x, Round 4800**										**DataDet-YOLO26x, Round 5400**									
Precision	99.0	93.3	98.0	80.2	99.7	98.6	99.8	98.0	95.7	95.8	98.0	96.4	97.8	78.9	99.5	97.2	99.9	98.0	96.3	95.8	97.9	95.3	91.1	74.1	99.3	99.4	99.9	97.5	94.0	94.3
Recall	92.2	89.8	97.3	98.0	85.8	98.3	93.9	88.2	81.5	91.7	97.2	85.7	97.3	98.0	87.0	98.3	92.4	88.8	80.2	91.7	92.5	89.6	98.1	97.9	93.2	97.9	94.3	87.3	73.4	91.6
F1-score	95.5	91.5	97.7	88.2	92.2	98.5	96.8	92.8	88.0	93.5	97.6	90.7	97.6	87.4	92.8	97.8	96.0	93.2	87.5	93.4	95.1	92.3	94.5	84.3	96.2	98.6	97.0	92.1	82.5	92.5
FPR	0.1	0.8	0.2	3.0	0.0	0.2	0.0	0.2	0.5	0.6	0.2	0.4	0.3	3.3	0.1	0.3	0.0	0.2	0.4	0.6	0.2	0.6	1.2	4.3	0.1	0.1	0.0	0.3	0.6	0.8
	DataDet-YOLO26x, Round 6000										**DataDet-YOLO26x, Round 6600**										**DataDet-YOLO26x, Round 7200**									
Precision	98.7	97.5	95.2	82.9	99.3	99.6	99.9	97.8	95.1	96.2	98.5	96.8	96.9	83.7	99.5	99.3	99.8	98.1	93.2	96.2	97.6	95.9	92.4	80.9	99.4	97.7	99.6	97.9	94.1	95.0
Recall	95.3	90.1	97.7	97.7	88.6	98.1	93.3	85.6	86.6	92.6	93.2	90.5	97.8	98.2	88.9	98.1	94.3	88.5	87.2	93.0	94.1	87.2	97.9	98.1	90.5	98.1	93.9	86.7	84.5	92.3
F1-score	97.0	93.6	96.4	89.7	93.6	98.9	96.5	91.3	90.6	94.2	95.8	93.6	97.4	90.4	93.9	98.7	97.0	93.0	90.1	94.4	95.8	91.4	95.1	88.7	94.8	97.9	96.7	91.9	89.0	93.5
FPR	0.2	0.3	0.6	2.5	0.1	0.0	0.0	0.2	0.6	0.5	0.2	0.4	0.4	2.4	0.1	0.1	0.0	0.2	0.8	0.5	0.3	0.5	1.0	2.9	0.1	0.3	0.0	0.2	0.7	0.7
	DataDet-YOLO26x, Round 7800										**DataDet-YOLO26x, Round 8400**										**DataDet-YOLO26x, Round 9000**									
Precision	99.2	97.8	94.0	84.1	99.6	99.1	99.9	98.0	96.2	96.4	98.0	96.9	96.5	82.0	99.4	98.5	99.9	98.4	96.4	96.2	95.7	87.9	94.0	80.8	98.8	99.0	99.9	98.2	93.8	94.2
Recall	96.8	91.1	98.0	97.8	88.9	98.4	93.3	88.3	88.4	93.5	96.2	90.5	97.9	98.2	90.5	98.4	94.1	88.2	84.0	93.1	89.8	89.3	97.0	98.2	95.0	97.8	89.6	86.8	80.0	91.5
F1-score	98.0	94.3	95.9	90.5	94.0	98.7	96.5	92.9	92.1	94.8	97.1	93.6	97.2	89.4	94.8	98.4	96.9	93.0	89.8	94.5	92.7	88.6	95.5	88.6	96.8	98.4	94.5	92.1	86.4	92.6
FPR	0.1	0.3	0.8	2.3	0.0	0.1	0.0	0.2	0.4	0.5	0.2	0.4	0.4	2.7	0.1	0.2	0.0	0.2	0.4	0.5	0.5	1.5	0.8	2.9	0.1	0.1	0.0	0.2	0.7	0.8

TABLE XXIII: Per-class (C1–C9) and average performance of DataDet-YOLO26x across 15 training rounds on DMD under the **RGB** training and **RGB** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	DataDet-YOLO26n, Round 600										**DataDet-YOLO26n, Round 1200**										**DataDet-YOLO26n, Round 1800**									
Precision	75.8	65.8	80.6	68.1	95.8	58.5	96.6	97.0	89.1	80.8	79.5	80.6	84.6	74.9	97.4	68.9	97.4	96.8	82.1	84.7	91.1	57.6	86.8	65.6	97.7	83.8	99.6	98.0	89.4	85.5
Recall	29.0	30.2	91.4	92.0	81.7	97.6	96.1	67.1	54.4	71.1	47.5	45.0	91.2	91.6	79.7	95.3	96.9	71.4	68.3	76.3	39.5	59.2	91.2	92.0	83.5	95.8	96.8	73.4	50.0	75.7
F1-score	42.0	41.4	85.7	78.3	88.2	73.1	96.4	79.4	67.5	72.4	59.5	57.8	87.8	82.4	87.7	80.0	97.1	82.2	74.6	78.8	55.1	58.4	88.9	76.6	90.1	89.4	98.2	83.9	64.2	78.3
FPR	1.2	2.0	2.7	5.4	0.4	8.7	0.4	0.3	0.8	2.4	1.5	1.3	2.1	3.8	0.3	5.4	0.3	0.3	1.9	1.9	0.5	5.4	1.7	6.0	0.2	2.3	0.0	0.2	0.7	1.9
	DataDet-YOLO26n, Round 2400										**DataDet-YOLO26n, Round 3000**										**DataDet-YOLO26n, Round 3600**									
Precision	84.6	82.6	72.6	70.3	99.0	68.4	99.1	97.6	88.4	84.7	88.5	74.6	76.5	73.0	97.9	81.5	99.4	97.4	86.4	86.1	88.8	81.3	77.4	80.0	95.2	79.8	98.3	98.3	87.5	87.4
Recall	55.9	45.1	91.5	91.2	79.1	97.1	96.7	58.9	59.0	74.9	59.1	52.8	92.0	91.6	80.6	97.5	96.7	60.4	60.2	76.8	63.5	53.7	91.5	91.7	80.6	96.8	96.7	58.3	73.8	78.5
F1-score	67.4	58.3	81.0	79.4	87.9	80.3	97.9	73.4	70.7	77.4	70.9	61.9	83.5	81.3	88.5	88.8	98.0	74.6	71.0	79.8	74.0	64.6	83.9	85.5	87.3	87.5	97.5	73.2	80.1	81.5
FPR	1.3	1.2	4.3	4.8	0.1	5.6	0.1	0.2	1.0	2.1	1.0	2.2	3.5	4.2	0.2	2.8	0.1	0.2	1.2	1.7	1.0	1.5	3.3	2.9	0.5	3.1	0.2	0.1	1.3	1.6
	DataDet-YOLO26n, Round 4200										**DataDet-YOLO26n, Round 4800**										**DataDet-YOLO26n, Round 5400**									
Precision	91.2	68.3	78.7	66.1	98.7	86.1	98.2	98.1	88.6	86.0	92.4	77.9	81.8	74.4	99.0	81.5	99.4	96.4	88.9	88.0	84.3	85.8	80.2	75.3	98.8	74.2	98.7	97.5	89.7	87.2
Recall	55.0	56.8	91.5	91.8	81.0	91.9	96.7	65.1	55.3	76.1	69.5	52.1	91.8	91.9	81.6	94.0	96.7	68.4	67.5	79.3	66.5	48.5	92.1	91.7	84.6	94.3	97.5	67.6	66.0	78.8
F1-score	68.6	62.0	84.6	76.9	89.0	88.9	97.4	78.2	68.1	79.3	79.3	62.5	86.5	82.2	89.5	87.3	98.0	80.0	76.7	82.5	74.4	62.0	85.8	82.7	91.2	83.0	98.1	79.9	76.0	81.4
FPR	0.7	3.3	3.1	5.9	0.1	1.9	0.2	0.2	0.9	1.8	0.7	1.8	2.6	4.0	0.1	2.7	0.1	0.3	1.1	1.5	1.6	1.0	2.8	3.8	0.1	4.1	0.2	0.2	0.9	1.6
	DataDet-YOLO26n, Round 6000										**DataDet-YOLO26n, Round –**										**DataDet-YOLO26n, Round –**									
Precision	90.7	72.7	86.1	80.0	98.0	81.0	99.9	97.7	88.3	88.3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	62.1	54.0	91.6	92.1	82.8	94.9	96.0	72.5	72.2	79.8	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	73.7	62.0	88.8	85.6	89.8	87.4	97.9	83.2	79.4	83.1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.8	2.5	1.8	2.9	0.2	2.8	0.0	0.2	1.2	1.4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XXIV: Per-class (C1–C9) and average performance of DataDet-YOLO26n across 10 training rounds on DMD under the **IR** training and **IR** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	DataDet-YOLO26s, Round 600										**DataDet-YOLO26s, Round 1200**										**DataDet-YOLO26s, Round 1800**									
Precision	88.0	90.0	86.3	71.6	95.3	77.3	99.4	97.9	89.3	88.3	80.8	90.8	88.2	70.8	98.5	87.2	99.4	97.4	88.0	89.0	91.8	78.6	83.6	74.3	99.1	76.8	99.4	97.4	88.2	87.7
Recall	71.9	42.2	91.5	91.4	82.0	97.2	96.6	70.7	64.4	78.6	72.5	59.4	91.1	90.7	84.1	95.3	96.8	74.1	59.1	80.3	61.3	50.3	91.1	92.3	80.6	97.6	96.8	67.6	67.7	78.4
F1-score	79.2	57.5	88.9	80.3	88.1	86.1	97.9	82.1	74.8	81.6	76.4	71.8	89.6	79.6	90.7	91.1	98.1	84.2	70.7	83.6	73.5	61.4	87.2	82.3	88.9	85.9	98.1	79.8	76.6	81.5
FPR	1.2	0.6	1.8	4.5	0.5	3.6	0.1	0.2	1.0	1.5	2.2	0.8	1.5	4.7	0.2	1.7	0.1	0.2	1.0	1.4	0.7	1.7	2.2	4.0	0.1	3.7	0.1	0.2	1.1	1.5
	DataDet-YOLO26s, Round 2400										**DataDet-YOLO26s, Round 3000**										**DataDet-YOLO26s, Round 3600**									
Precision	91.7	90.3	82.6	78.2	98.5	75.4	99.5	97.7	87.5	89.0	92.9	79.0	82.2	75.6	99.2	85.7	99.9	97.5	84.8	88.5	94.5	85.6	86.6	75.4	98.9	79.2	99.9	97.5	85.9	89.3
Recall	66.8	52.8	91.4	92.3	81.2	97.0	95.9	65.1	74.3	79.6	66.8	60.4	91.3	91.6	81.7	93.8	96.9	65.6	68.6	79.6	67.6	61.5	91.0	91.7	83.2	96.8	96.1	73.2	67.6	81.0
F1-score	77.3	66.6	86.8	84.6	89.0	84.8	97.7	78.1	80.3	82.8	77.7	68.5	86.5	82.8	89.6	89.6	98.4	78.4	75.8	83.0	78.8	71.6	88.7	82.8	90.4	87.1	98.0	83.6	75.7	84.1
FPR	0.8	0.7	2.4	3.2	0.2	4.0	0.1	0.2	1.3	1.4	0.6	2.0	2.5	3.7	0.1	2.0	0.0	0.2	1.5	1.4	0.5	1.3	1.8	3.7	0.1	3.2	0.0	0.2	1.4	1.4
	DataDet-YOLO26s, Round 4200										**DataDet-YOLO26s, Round 4800**										**DataDet-YOLO26s, Round 5400**									
Precision	94.7	85.8	82.0	78.6	98.6	82.0	99.9	98.4	89.7	90.0	93.4	84.9	86.2	75.6	99.3	82.8	99.9	97.2	86.1	89.5	91.1	94.8	87.2	73.4	99.2	86.7	99.6	97.8	87.7	90.8
Recall	75.4	61.6	91.6	91.6	83.8	93.3	96.2	65.5	74.4	81.5	69.7	62.6	91.0	92.1	80.6	96.8	96.0	72.7	68.1	81.1	81.9	56.0	91.4	92.1	82.6	96.7	96.6	74.3	67.3	82.1
F1-score	83.9	71.7	86.5	84.6	90.6	87.3	98.0	78.7	81.4	84.7	79.8	72.1	88.5	83.0	89.0	89.2	97.9	83.2	76.1	84.3	86.3	70.4	89.2	81.7	90.2	91.4	98.1	84.5	76.2	85.3
FPR	0.5	1.3	2.5	3.1	0.1	2.6	0.0	0.1	1.1	1.3	0.6	1.4	1.8	3.7	0.1	2.5	0.0	0.3	1.4	1.3	1.0	0.4	1.7	4.2	0.1	1.9	0.0	0.2	1.2	1.2
	DataDet-YOLO26s, Round 6000										**DataDet-YOLO26s, Round –**										**DataDet-YOLO26s, Round –**									
Precision	92.2	92.8	82.9	75.4	99.0	77.6	99.7	97.8	86.6	89.3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	75.3	53.1	91.9	91.6	82.7	96.2	96.0	69.0	69.8	80.6	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	82.9	67.5	87.2	82.7	90.1	85.9	97.8	80.9	77.3	83.6	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.8	0.5	2.4	3.7	0.1	3.5	0.0	0.2	1.3	1.4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XXV: Per-class (C1–C9) and average performance of DataDet-YOLO26s across 10 training rounds on DMD under the **IR** training and **IR** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	DataDet-YOLO26m, Round 600										**DataDet-YOLO26m, Round 1200**										**DataDet-YOLO26m, Round 1800**									
Precision	86.2	78.2	90.0	72.1	97.6	77.2	99.7	97.2	87.0	87.2	89.8	88.4	91.1	70.5	99.1	85.5	99.7	97.6	88.0	90.0	93.6	68.9	89.3	70.2	99.2	83.8	99.1	98.0	86.9	87.7
Recall	59.9	41.3	91.4	89.7	75.8	97.3	96.5	72.4	66.4	76.7	68.3	57.1	90.8	91.1	77.4	96.3	96.8	75.5	61.7	79.4	53.8	49.0	91.3	92.3	77.8	97.1	96.5	73.7	59.8	76.8
F1-score	70.7	54.0	90.7	80.0	85.3	86.1	98.1	83.0	75.3	80.3	77.6	69.4	90.9	79.5	86.9	90.6	98.2	85.1	72.5	83.4	68.3	57.2	90.3	79.8	87.2	90.0	97.8	84.1	70.8	80.6
FPR	1.2	1.4	1.3	4.3	0.2	3.6	0.0	0.3	1.2	1.5	1.0	0.9	1.1	4.8	0.1	2.0	0.0	0.2	1.0	1.2	0.5	2.8	1.4	4.9	0.1	2.4	0.1	0.2	1.1	1.5
	DataDet-YOLO26m, Round 2400										**DataDet-YOLO26m, Round 3000**										**DataDet-YOLO26m, Round 3600**									
Precision	89.8	94.2	85.0	76.6	98.6	78.1	98.2	97.4	89.0	89.6	92.7	91.3	86.9	78.0	98.4	84.8	99.4	97.4	85.0	90.4	89.1	94.5	87.6	75.0	98.3	83.1	99.8	97.8	87.9	90.3
Recall	74.6	43.7	91.3	91.4	77.6	97.3	96.1	69.8	68.4	78.9	75.8	52.8	91.3	92.0	76.1	97.2	96.6	70.4	71.6	80.4	80.5	44.8	91.2	92.0	78.8	97.1	96.8	72.5	65.8	79.9
F1-score	81.5	59.7	88.0	83.3	86.9	86.6	97.1	81.3	77.4	82.4	83.4	66.9	89.0	84.4	85.8	90.6	98.0	81.7	77.8	84.2	84.6	60.8	89.3	82.7	87.5	89.5	98.3	83.2	75.2	83.5
FPR	1.1	0.3	2.0	3.5	0.1	3.4	0.2	0.2	1.1	1.3	0.7	0.6	1.7	3.3	0.2	2.2	0.1	0.2	1.6	1.2	1.2	0.3	1.6	3.8	0.2	2.5	0.0	0.2	1.1	1.2
	DataDet-YOLO26m, Round 4200										**DataDet-YOLO26m, Round 4800**										**DataDet-YOLO26m, Round 5400**									
Precision	93.8	92.2	89.9	80.7	99.2	86.9	99.8	98.1	93.4	92.7	94.7	86.6	86.3	77.2	98.7	83.0	99.7	97.7	89.3	90.4	91.8	93.7	90.4	78.5	98.1	86.9	99.4	97.7	92.5	92.1
Recall	78.8	55.2	91.4	92.0	77.2	97.3	96.9	74.4	77.9	82.4	75.2	50.1	91.4	91.1	77.5	97.3	96.6	70.1	73.0	80.3	78.1	55.4	91.6	91.9	80.9	97.2	96.9	77.6	74.7	82.7
F1-score	85.6	69.0	90.7	86.0	86.8	91.8	98.4	84.6	84.9	86.4	83.8	63.5	88.8	83.6	86.8	89.6	98.1	81.7	80.3	84.0	84.4	69.6	91.0	84.7	88.7	91.8	98.1	86.5	82.7	86.4
FPR	0.7	0.6	1.3	2.7	0.1	1.8	0.0	0.2	0.7	0.9	0.5	1.0	1.8	3.4	0.1	2.5	0.0	0.2	1.1	1.2	0.9	0.5	1.2	3.1	0.2	1.8	0.1	0.2	0.8	1.0
	DataDet-YOLO26m, Round 6000										**DataDet-YOLO26m, Round –**										**DataDet-YOLO26m, Round –**									
Precision	92.9	89.5	87.7	72.3	99.4	85.8	99.0	97.7	90.0	90.5	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	75.5	54.4	91.7	91.4	77.9	97.0	96.4	73.0	65.9	80.4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	83.3	67.6	89.7	80.7	87.3	91.0	97.7	83.6	76.1	84.1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.7	0.8	1.6	4.4	0.1	2.0	0.1	0.2	0.9	1.2	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XXVI: Per-class (C1–C9) and average performance of DataDet-YOLO26m across 10 training rounds on DMD under the **IR** training and **IR** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	DataDet-YOLO26l, Round 600										**DataDet-YOLO26l, Round 1200**										**DataDet-YOLO26l, Round 1800**									
Precision	87.0	89.4	85.1	68.9	98.9	78.7	99.9	97.5	84.0	87.7	87.1	91.1	87.1	70.4	98.5	82.8	99.6	97.3	88.5	89.2	91.5	79.7	88.7	70.9	99.0	88.1	99.7	97.5	88.0	89.2
Recall	66.5	44.9	91.3	91.1	78.0	97.2	96.2	68.9	60.1	77.1	71.3	51.9	91.0	91.3	77.8	97.3	97.1	71.1	60.4	78.8	64.9	54.3	91.0	91.8	79.8	97.0	97.2	72.4	63.8	79.1
F1-score	75.4	59.8	88.1	78.4	87.2	87.0	98.0	80.8	70.1	80.5	78.4	66.2	89.0	79.5	86.9	89.4	98.4	82.2	71.8	82.4	76.0	64.6	89.8	80.0	88.4	92.3	98.5	83.1	74.0	83.0
FPR	1.2	0.7	2.0	5.1	0.1	3.3	0.0	0.2	1.4	1.6	1.3	0.6	1.7	4.8	0.1	2.5	0.0	0.2	1.0	1.4	0.8	1.7	1.5	4.7	0.1	1.6	0.0	0.2	1.1	1.3
	DataDet-YOLO26l, Round 2400										**DataDet-YOLO26l, Round 3000**										**DataDet-YOLO26l, Round 3600**									
Precision	92.8	88.8	84.9	76.8	98.7	85.2	99.1	97.0	89.4	90.3	91.5	89.6	85.7	78.9	98.9	78.7	99.7	97.4	85.1	89.5	86.3	87.6	87.1	82.3	98.8	85.7	99.8	97.7	85.6	90.1
Recall	71.8	56.4	91.4	92.1	78.0	96.6	96.6	69.8	71.9	80.5	76.3	41.1	91.2	92.3	81.5	97.0	96.9	71.6	73.4	80.2	73.8	49.0	91.3	92.0	79.9	97.0	97.2	72.0	76.5	81.0
F1-score	80.9	68.9	88.1	83.7	87.2	90.5	97.8	81.2	79.7	84.2	83.2	56.4	88.4	85.1	89.4	86.9	98.3	82.6	78.8	83.2	79.6	62.8	89.1	86.9	88.4	91.0	98.5	82.9	80.8	84.4
FPR	0.7	0.9	2.0	3.5	0.1	2.1	0.1	0.3	1.1	1.2	0.9	0.6	1.9	3.1	0.1	3.3	0.0	0.2	1.6	1.3	1.5	0.9	1.7	2.5	0.1	2.0	0.0	0.2	1.6	1.2
	DataDet-YOLO26l, Round 4200										**DataDet-YOLO26l, Round 4800**										**DataDet-YOLO26l, Round 5400**									
Precision	93.0	88.3	89.2	74.5	98.7	83.0	99.9	98.2	92.0	90.8	93.8	92.7	88.1	77.9	98.9	83.2	99.5	97.3	89.0	91.2	94.7	91.2	91.9	78.2	99.1	89.0	99.7	97.7	90.3	92.4
Recall	78.2	52.8	91.4	91.6	79.7	96.9	96.3	73.8	69.7	81.2	77.4	54.0	91.5	92.2	81.1	97.4	97.0	72.7	71.8	81.7	79.3	59.9	91.4	92.1	79.3	97.2	96.7	77.4	72.9	82.9
F1-score	85.0	66.1	90.3	82.2	88.2	89.5	98.1	84.3	79.3	84.8	84.8	68.3	89.8	84.5	89.1	89.8	98.3	83.3	79.5	85.2	86.3	72.3	91.6	84.6	88.1	92.9	98.2	86.4	80.7	86.8
FPR	0.7	0.9	1.4	3.9	0.1	2.5	0.0	0.2	0.8	1.2	0.6	0.5	1.6	3.3	0.1	2.5	0.1	0.2	1.1	1.1	0.6	0.7	1.0	3.2	0.1	1.5	0.0	0.2	1.0	0.9
	DataDet-YOLO26l, Round 6000										**DataDet-YOLO26l, Round –**										**DataDet-YOLO26l, Round –**									
Precision	93.2	93.6	86.8	75.8	98.4	85.7	99.7	97.6	88.8	91.1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	77.2	58.2	91.4	92.1	81.2	97.2	96.6	73.3	70.3	81.9	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	84.4	71.7	89.0	83.2	89.0	91.1	98.1	83.7	78.5	85.4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.7	0.5	1.7	3.7	0.2	2.0	0.0	0.2	1.1	1.1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XXVII: Per-class (C1–C9) and average performance of DataDet-YOLO26l across 10 training rounds on DMD under the **IR** training and **IR** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.

Metric (%)	Classes									Avg.	Classes									Avg.	Classes									Avg.
	C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9		C1	C2	C3	C4	C5	C6	C7	C8	C9	
	DataDet-YOLO26x, Round 600										**DataDet-YOLO26x, Round 1200**										**DataDet-YOLO26x, Round 1800**									
Precision	92.5	78.5	92.1	70.4	99.4	78.8	99.8	96.5	79.9	87.5	85.8	92.2	88.7	71.1	98.9	88.8	99.8	97.3	88.3	90.1	90.9	80.3	87.2	74.9	99.3	89.9	99.4	96.8	85.7	89.4
Recall	61.1	42.8	91.1	88.9	73.4	97.1	96.9	76.9	59.7	76.4	75.8	51.0	91.1	91.5	79.3	97.0	97.0	73.5	61.0	79.7	64.0	56.1	91.5	92.1	78.4	97.0	97.0	72.5	69.5	79.8
F1-score	73.6	55.4	91.6	78.6	84.4	87.0	98.3	85.6	68.3	80.3	80.5	65.7	89.9	80.0	88.0	92.7	98.4	83.8	72.2	83.5	75.1	66.0	89.3	82.6	87.6	93.3	98.2	82.9	76.7	83.5
FPR	0.6	1.5	1.0	4.7	0.1	3.3	0.0	0.3	1.9	1.5	1.6	0.5	1.5	4.7	0.1	1.5	0.0	0.3	1.0	1.2	0.8	1.7	1.7	3.8	0.1	1.4	0.1	0.3	1.5	1.3
	DataDet-YOLO26x, Round 2400										**DataDet-YOLO26x, Round 3000**										**DataDet-YOLO26x, Round 3600**									
Precision	91.3	91.7	84.0	76.3	98.9	84.6	99.9	97.4	84.5	89.8	91.1	85.7	89.1	75.4	98.5	86.2	99.9	97.5	87.8	90.1	87.8	84.5	87.5	80.3	98.2	82.6	99.8	97.2	88.5	89.6
Recall	72.9	55.5	91.2	92.0	78.8	96.9	96.7	68.5	68.0	80.1	76.6	54.2	90.6	92.0	77.2	96.4	96.9	74.3	66.3	80.5	71.9	47.5	91.4	91.8	77.1	97.0	97.0	71.1	72.2	79.7
F1-score	81.1	69.2	87.4	83.4	87.7	90.4	98.3	80.4	75.4	83.7	83.2	66.4	89.9	82.9	86.6	91.0	98.4	84.3	75.6	84.2	79.1	60.8	89.4	85.6	86.4	89.2	98.4	82.1	79.5	83.4
FPR	0.9	0.6	2.2	3.6	0.1	2.2	0.0	0.2	1.6	1.3	0.9	1.1	1.4	3.8	0.1	1.9	0.0	0.2	1.1	1.2	1.2	1.1	1.6	2.8	0.2	2.6	0.0	0.3	1.2	1.2
	DataDet-YOLO26x, Round 4200										**DataDet-YOLO26x, Round 4800**										**DataDet-YOLO26x, Round 5400**									
Precision	93.0	90.5	91.4	79.2	99.0	88.1	99.8	97.9	92.4	92.4	93.3	91.7	82.9	79.4	99.1	83.9	99.8	97.5	90.4	90.9	94.0	94.1	89.2	77.6	99.0	89.3	99.8	97.8	89.7	92.3
Recall	78.9	60.1	91.3	92.0	78.2	97.1	97.0	76.3	74.6	82.8	80.7	53.6	91.6	91.9	82.7	97.3	97.1	67.7	74.4	81.9	81.1	61.5	91.4	92.2	80.1	97.2	96.5	75.2	72.3	83.0
F1-score	85.4	72.2	91.4	85.1	87.4	92.4	98.4	85.8	82.6	86.7	86.5	67.6	87.0	85.2	90.2	90.1	98.5	79.9	81.6	85.2	87.1	74.4	90.3	84.2	88.6	93.1	98.1	85.0	80.1	86.7
FPR	0.7	0.8	1.1	3.0	0.1	1.6	0.0	0.2	0.8	0.9	0.7	0.6	2.4	3.0	0.1	2.3	0.0	0.2	1.0	1.1	0.7	0.5	1.4	3.3	0.1	1.5	0.0	0.2	1.0	1.0
	DataDet-YOLO26x, Round 6000										**DataDet-YOLO26x, Round –**										**DataDet-YOLO26x, Round –**									
Precision	94.1	87.0	91.3	78.9	99.1	86.1	99.4	97.6	89.2	91.4	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Recall	74.7	60.4	91.4	91.9	80.7	97.2	96.9	77.1	74.2	82.7	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
F1-score	83.3	71.3	91.4	84.9	89.0	91.3	98.1	86.2	81.0	86.3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
FPR	0.6	1.1	1.1	3.1	0.1	2.0	0.1	0.2	1.1	1.0	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE XXVIII: Per-class (C1–C9) and average performance of DataDet-YOLO26x across 10 training rounds on DMD under the **IR** training and **IR** test setting. C1–C9 denote `drinking`, `hair_and_makeup`, `hold_on_the_wheel_left`, `hold_on_the_wheel_right`, `phonecall_left`, `phonecall_right`, `radio`, `texting_left`, and `texting_right`, respectively.